

After Collusion: Cartel Endings and the Reorientation of Minority Ownership

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Abstract

We follow firms found by the European Commission to have participated in cartels, linking decisions issued between 2001 and 2022 to firm accounts and worldwide transaction records, and ask how the end of explicit coordination changes the ownership links they form. In the first three years after the recorded end of the infringement, a member's annual probability of acquiring a horizontal minority stake rises by 2.8 percentage points relative to matched comparison firms. Against a pre-ending rate of 1.8 percent, the annual probability more than doubles. Full acquisitions and overall minority investment show no comparable change, we detect no corresponding response on the target side, and the same firms' minority investments shift from non-horizontal toward horizontal targets. The acquired firms are not former cartel partners, and most operate outside the industry named in the decision. Cartel endings are thus followed by a broad reorientation of partial ownership toward competitors, through transactions that leave every target a separate company and largely escape control-based merger review.

Keywords minority shareholdings, partial ownership, cartels, collusion, merger control.

JEL codes G34, K21, L41.

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1 Introduction

In June 2025, the European Commission fined Delivery Hero and Glovo EUR 329 million for dividing markets, exchanging commercially sensitive information, and agreeing not to hire one another's employees.¹ The case was unusual because the relationship had begun with a minority investment. Delivery Hero's stake in Glovo created financial ties, governance rights, and repeated contact between two firms that remained separate competitors, and the Commission concluded that these links helped sustain a cartel. It was the first time the Commission sanctioned the anticompetitive use of a minority shareholding in a rival.

The case illustrates a familiar direction of causality. Ownership can change how firms compete and, in some settings, help them coordinate. This paper asks what happens in the opposite direction. When explicit coordination ends, do former cartel members change the ownership links they form?

The question matters because the end of a cartel does not erase the relationships that sustained it. Former members know one another's products, customers, capacity, and managers, and they face a renewed problem. Actions that steal business from a rival once again become privately attractive, even when they reduce the joint profit of the firms that used to coordinate. A full merger is one way to internalize that tension. A minority stake offers a less complete alternative, giving the acquirer a financial interest in another firm while both companies stay in the market. Such stakes can carry voting rights, board representation, or access to information, yet because they rarely confer control, most escape the notification requirements of merger review—much as the below-threshold mergers studied by Wollmann (2019) escape premerger scrutiny.²

Minority ownership need not substitute for collusion. A stake may instead make a cartel easier to sustain, by reducing the gains from cheating or by improving monitoring (Malueg, 1992; Reitman, 1994; Gilo et al., 2006). Nor is a cartel ending a laboratory event. Infringements can end amid leniency applications, investigations, management changes, and wider corporate restructuring, forces that can move ownership in either direction. Whether former cartelists acquire stakes in horizontally related firms is an empirical question.

We answer it by linking European Commission cartel decisions to Bureau van Dijk firm records and worldwide transaction data. The decisions identify the firms that participated in each cartel and the final year for which the infringement was established; the transaction data then follow the same firms as acquirers and as targets. Our main sample contains 235 named members from 67 cartels whose recorded endings span 2001 to 2015. For every member and

¹European Commission, press release IP/25/1356, June 2025, https://ec.europa.eu/commission/presscorner/detail/en/ip_25_1356. Announcing the decision, Executive Vice-President Teresa Ribera noted that “owning a stake in a competitor is not illegal in itself,” but that such stakes can be “used to gain insight information and influence decisions in ways that can harm competition” (press conference, 2 June 2025).

²Wollmann (2019) shows that when an amendment to the Hart-Scott-Rodino Act exempted smaller United States mergers from premerger notification, horizontal deals among the newly exempt transactions rose sharply. Non-controlling minority stakes sit outside control-based review for a different legal reason, but the enforcement consequence is similar, and the transactions proceed largely unexamined.

cartel event, we construct a comparison group of firms with similar acquisition histories, financial characteristics, and broad industry exposure before the ending, and we follow both groups from four years before through four years after the recorded end of the infringement.

The change is visible before any event-study estimator is applied. Members and their comparison firms enter the event window with similar acquisition records, and their paths separate only once the cartel ends. During the first three full post-ending years, a member's annual probability of acquiring a horizontal minority stake rises by 2.8 percentage points relative to its comparison group. The rate before the ending is 1.8 percent, so the probability more than doubles. More members make such acquisitions after the ending—an extensive-margin result—and the annual effect fades in year four.

This is not a general acquisition wave. Full acquisitions show no parallel change, the probability of making any minority investment stays close to its earlier path, and we detect no corresponding response on the target side. Instead, the direction of investment shifts. For the same firms and comparison groups, a horizontal-minus-non-horizontal measure turns sharply positive after the cartel ending.

The transactions themselves do not simply reproduce the old agreement. None of the targets is a former member of the same cartel, and most lie outside the four-digit industry named in the decision. They range from E.ON assembling minority positions in Central European gas utilities within a year of the end of the E.ON–GDF market-sharing conduct, to Robert Bosch joining the financing round of an organic-solar-film producer three years after the spark-plug cartel ended (Section 6.4). The data show a turn toward horizontally related firms across the acquirer's businesses, broader than the replacement of a former cartel partner.

This pattern is difficult to attribute to chance or to background conditions. Shifting the event two, three, or four years backward produces nothing similar, and when event time is organized around the much later Commission decision, the response has already begun. Nor is the pattern easy to generate among other firms. Assigning the event to comparison firms never named in a cartel, or flipping entire cartels' contributions at random, yields empirical p -values of 0.001 and 0.009, and the average firm in a cartel-affected industry shows no comparable change.

Firms can align their interests with those of rivals in more than one way. Explicit coordination is one such way, and partial ownership is another. We contribute to the empirical work on cartels by documenting how firms' use of the second changes when the first ends—an adjustment margin that has gone unmeasured. To our knowledge, these are the first estimates of how cartel members' ownership links respond to the end of their own agreement. Cartel episodes affect prices, investment, and innovation: Kang (2025) shows that colluding firms' innovation reverts to pre-collusion levels once the cartel dissolves. Cartel breakdown can also be followed by consolidation (Davies et al., 2015; Dong et al., 2019) and by changes in public disclosure (Bourveau et al., 2020). Those margins involve control or communication. The margin we document alters incentives while leaving every firm in the market, invisible

to firm counts and to standard market-share concentration measures.

Research on partial ownership and collusion has examined how ownership affects competition, deviation incentives, information exchange, and cartel stability (Reynolds and Snapp, 1986; O'Brien and Salop, 2000; Gilo et al., 2006; Nain and Wang, 2018; Abraham et al., 2025), and Heim et al. (2022) show that horizontal minority acquisitions increase when leniency programs raise the expected cost of explicit collusion. We contribute by examining the next stage of the same relationship: not whether ownership changes the feasibility of collusion, but how the realized end of a firm's own agreement changes the ownership links it forms, a reorientation that reaches across the member's businesses rather than only the market the cartel had covered.

Finally, the results locate the change in the firms themselves rather than in market conditions. The average firm in a cartel-affected industry does not respond.

The evidence has limits. The recorded ending is an institutional boundary rather than a randomized withdrawal of coordination; it can coincide with an investigation, a leniency application, or a broader change in corporate strategy. The transaction data do not reveal every voting arrangement attached to a stake, and we cannot show that any particular deal replaced collusion. The data establish something narrower and new. Once explicit coordination ends, former cartel members become more likely to build horizontal ownership links through transactions recorded as minority stakes rather than full acquisitions, leaving each target in place as a separate company.

The remainder of the paper proceeds as follows. Section 2 explains how cartel endings and minority investments are recorded. Section 3 sets out why ownership may become more or less attractive when collusion ends, and what different responses would look like in the data. Sections 4 and 5 introduce the data and the empirical design. Section 6 presents the results, Section 7 discusses their interpretation, and Section 8 concludes.

2 Cartel Endings and Minority Ownership

2.1 Cartel endings and enforcement timing

European Commission decisions reconstruct an infringement from legal and economic evidence. They identify the undertakings that participated, the products or services covered by the agreement, the conduct at issue, and the period for which the Commission established a violation. Price fixing appears in about 85 percent of the cases; market or customer allocation and exchanges of commercially sensitive information each appear in roughly half, quantity restrictions in a quarter, and bid coordination in six percent. Many cases combine several forms of conduct.

The affected activities range from gas distribution and automotive components to food packaging and interest rate derivatives, and the arrangements themselves took very different organizational forms. Prestressing-steel producers coordinated through a structure known

as Club Zurich, later Club Europe, with regional branches including Club Italia and Club España; candle-wax producers met in the “Blauer Salon”; and foreign-exchange traders used a chatroom called “Essex Express ‘n the Jimmy.”³ The eventual decisions imposed substantial penalties—the median case fine was EUR 97 million, and the largest, in the Trucks case, EUR 3.8 billion—but typically came years after the conduct had ended.

Several dates are attached to such a case. The first and last years of the proven infringement describe the period of conduct, while an inspection, immunity application, statement of objections, settlement, and final decision each mark a stage of the enforcement procedure.⁴ We date the event by the final calendar year of the proven infringement: the last year in which the Commission established that the cartel operated. This date is closer to the economic relationship of interest than the decision year, though it is not an exact timestamp for a sudden transition to competition. One member may leave before another, parts of an agreement can unravel gradually, and an immunity application or dawn raid may itself bring the conduct to an end.⁵ The literature on cartel duration shows that many cartels end under enforcement pressure rather than on their own (Levenstein and Suslow, 2006, 2011). We treat the recorded date as the documented boundary of the infringement, and Section 5 explains how this reading shapes the identifying assumption.

Because the ending process differs across cases, we examine it directly. For all 67 cartels in the matched sample, we read the Commission decisions and related official records and record the first documented leniency or immunity application, complaint, inspection, or other authority action. Thirty-seven cartels ended before any such step is documented; the remaining thirty ended after enforcement had become visible or on the same date as the triggering action. In twenty of the thirty, the first documented step is a leniency or immunity application; inspections, complaints, and other authority actions account for the rest. The classification describes the order of events and does not assume that a cartel ending before visible enforcement was voluntary; Section 6.5 uses it to ask whether the ownership response depends on how the cartel ended.

The institutional setting offers an unusual empirical opportunity. A Commission decision identifies the firms that were connected by an illegal agreement and places an external date on the end of the proven conduct. Most corporate relationships come with neither piece of information. We can ask how the same named firms change their ownership choices around a common boundary, and compare those choices with firms that were never named in the cartel records.

³Cases AT.38344 (*Prestressing steel*), AT.39181 (*Candle waxes*), and AT.40135 (*Forex*). All three appear in the estimation sample.

⁴The decision is usually the most visible of these dates, but it often lies years after the conduct itself—in our sample, a median of five years after the last proven infringement year. A study of what firms do after collusion should not date the event by the publication of the legal decision.

⁵The E.ON–GDF case illustrates how much history can lie behind a single recorded ending. The firms had reserved the German and French gas markets for each other in side letters signed in 1975. They declared the letters void in 2004, but the Commission found that the infringement continued into 2005, the final infringement year recorded for the case (COMP/39.401, decision of 8 July 2009).

2.2 Minority stakes and the boundary of merger control

A full acquisition combines buyer and target under common control. A minority investment leaves the target as a separate undertaking while transferring part of its cash-flow rights to the acquirer, sometimes together with voting rights, board representation, access to information, or influence over important decisions. Firms use partial stakes for many benign purposes: to finance growth, share risk, support supply or research relationships, learn about a potential partner, and preserve the option of a later acquisition (Allen and Phillips, 2000; Fee et al., 2006; Ouimet, 2013; Liao, 2014; Bostan and Spatareanu, 2018). The same contractual flexibility matters for competition, for reasons that Section 3 develops.

The distinction from a merger also matters for enforcement. The EU Merger Regulation is built around transactions that confer control, so a minority investment that does not confer control generally falls outside the notification process.⁶ A former cartel member that buys a fraction of another firm consequently creates an economic link that EU merger control is not required to review in advance.⁷

The legal line between control and non-control cannot be recovered perfectly from transaction databases. A stake below 50 percent may confer joint control or substantial influence, while a larger financial interest can sometimes remain passive. Our data identify transactions reported as minority investments and exclude deals whose known final ownership share reaches 50 percent; when the exact percentage is missing, we rely on the provider's deal classification.⁸ The economic distinction that motivates the paper survives this measurement boundary. The observed transactions leave the target recorded as a separate company, so they create a financial link without the visible reduction in firm count that accompanies a merger. If former cartel members turn toward such stakes, post-cartel consolidation measures that track only acquisitions of control will miss part of the adjustment.

⁶The European Commission considered a broader system in its 2014 White Paper, motivated in part by the possibility that non-controlling shareholdings could weaken competition (European Commission, 2014). A later evaluation did not find sufficient support for the proposed general framework (European Commission, 2021), but the concern has returned. The draft merger guidelines published for consultation in April 2026 treat a non-controlling minority shareholding in a competitor as a possible source of harm—"financial interests, influence, or information flows that weaken the incentives of the merged firm and the competing firm to compete with each other"—and leave stakes below 5 percent aside only absent additional rights or links (European Commission, 2026, paras. 163–165). The draft governs the assessment of notified mergers; a stand-alone acquisition of a non-controlling stake still requires no notification. Other provisions of competition law can still apply when a stake is tied to anticompetitive conduct; Delivery Hero–Glovo is the clearest recent example.

⁷The gap is a feature of control-based regimes rather than of competition law generally. Germany and Austria review acquisitions of 25 percent or more, and the United Kingdom's "material influence" standard allowed its competition authority to order Ryanair to unwind a 29.8 percent stake in Aer Lingus. United States agencies can reach partial acquisitions under Section 7 of the Clayton Act, and the 2023 Merger Guidelines warn that a partial acquisition can lessen competition by giving the acquirer "the ability to influence the competitive conduct of the target firm," by "reducing the incentive of the acquiring firm to compete," or by providing "access to non-public, competitively sensitive information" (U.S. Department of Justice and Federal Trade Commission, 2023). See Organisation for Economic Co-operation and Development (OECD) (2009) for a survey across jurisdictions.

⁸We refer to minority-stake acquisitions rather than asserting that every transaction is legally non-controlling. Section 4.2 describes the data and the available ownership shares.

3 Ownership as a Substitute for Collusion?

When two rivals collude, they already take account of one another's actions through the agreement. A price cut, a capacity expansion, or entry into a customer's territory is no longer chosen independently. The cartel suppresses precisely the competitive actions through which one member would otherwise harm another.

A minority stake creates a different and more durable link. An acquirer that holds a share α of a rival's equity receives α of that rival's profit, and so bears part of the loss that its own aggressive actions impose—the standard unilateral effect of partial ownership (Reynolds and Snapp, 1986; Bresnahan and Salop, 1986; Farrell and Shapiro, 1990; O'Brien and Salop, 2000; Brito et al., 2014). The internalization is far from trivial at the stakes we observe. Among the post-ending transactions that report an ownership share, the median is just under 16 percent, and at that stake an acquirer absorbs roughly one-sixth of the profit its competition destroys at the target. While a cartel operates, explicit coordination already limits business stealing, so such a stake adds little. Once the agreement ends and firms again choose their actions independently, ownership can become more valuable as a way to soften the externality between them. In this account, partial ownership and collusion are substitutes, and the demand for ownership should rise when collusion ends.

The opposite relationship is also possible. A stake can reduce the gain from cheating, improve monitoring, or ease communication, in which case partial ownership helps sustain a cartel and is more attractive while coordination is still in place (Malueg, 1992; Reitman, 1994; Spagnolo, 1998; Gilo et al., 2006; de Haas and Paha, 2021). An investigation can further tighten financial constraints or make horizontal links legally more costly. Theory gives a sharp reason to look at ownership around a cartel ending, but it does not settle the sign.

If ownership does respond, where should the new stakes go? The narrowest version of the substitution account points to former co-cartelists and to rivals in the affected market, because these are the firms whose behavior was previously coordinated and whose renewed competition now threatens profit. A broader version is equally coherent *ex ante*. Cartel members are often diversified, operating in several markets at once, and the end of one agreement can prompt management to reassess competitive relationships across the whole portfolio—particularly where the same logic of profit internalization applies to rivals the firm was never allowed to approach collusively. Both versions predict horizontal stakes on the buyer side; they differ in the identity of the targets. We treat them as competing hypotheses and let the transaction destinations distinguish them, rather than building either into the definition of the outcome.

These considerations translate into empirical patterns. A reorganization of competitive relationships through ownership should appear on the buyer side and be directed toward horizontal firms; it need not occur within the final infringement year, because identifying a target, negotiating a stake, and completing a transaction take time. A general restructuring or financing shock, by contrast, should also move total minority investment or full acquisitions.

And the location of the targets—inside the affected market, or spread across the acquirer’s other businesses—separates the narrow from the broad version of the ownership response.

Our setting also differs from the common-ownership literature. Research on common ownership studies outside investors who hold stakes in several competitors (Azar et al., 2018; Lopez and Vives, 2019; Backus et al., 2021; Koch et al., 2021; Azar et al., 2022), and direct stakes held by operating firms have been shown to soften unilateral incentives or facilitate coordination (Dietzenbacher et al., 2000; Nain and Wang, 2018). Here, an operating company itself chooses to buy part of another firm, and the question is when that choice is made.

4 Data

The analysis combines three types of information that are rarely observed together. European Commission decisions identify cartel cases, their participants, and the period of proven conduct. Bureau van Dijk records connect those legal names to firms and provide accounts and industry classifications. Worldwide transaction records from Bureau van Dijk’s Zephyr database show when the same firms acquire, or become targets of, minority stakes.

4.1 Cartel cases and member firms

We begin with a database of European Commission cartel decisions. It records the products and conduct described in each case, the start and end of the infringement, the Commission decision year, and the undertakings reported as participants.⁹ The database contains 129 cartels and is current through decisions published in 2025.

Legal decisions name undertakings rather than stable firm identifiers, so a separate concordance links company names, aliases, and subsidiaries to Bureau van Dijk identifiers. A firm enters the treated population only when the Commission names it as a participant in the assigned case—firms are not treated merely because their industry was affected by a cartel. The event year g is the final calendar year of the proven infringement. A firm that participated in two eligible cartels can enter twice, with a different event date and a different comparison group; there are 246 firm–case events and 235 distinct members in the main sample, with eleven firms appearing twice. The results are similar when only the first or only the last event is retained.

The estimation requires the complete event window from $g - 4$ through $g + 4$, the pre-existing accounts used to construct comparison groups, and at least one suitable comparison firm. These requirements reduce the 1,717 named-member firm–case events in the source panel to 470 financially and temporally eligible events and, once at least one suitable comparison firm is required, to 246 matched events in 67 cartels; Appendix Table A.2 reports each step. The matched cartel endings span 2001 to 2015, an endpoint that reflects data support

⁹The database is compiled from the Commission’s public registry of competition cases (<https://competition-cases.ec.europa.eu/search?caseInstrument=AT&caseCartel=true>) and covers cartel cases with a final Commission decision; the earliest recorded conduct begins in 1971.

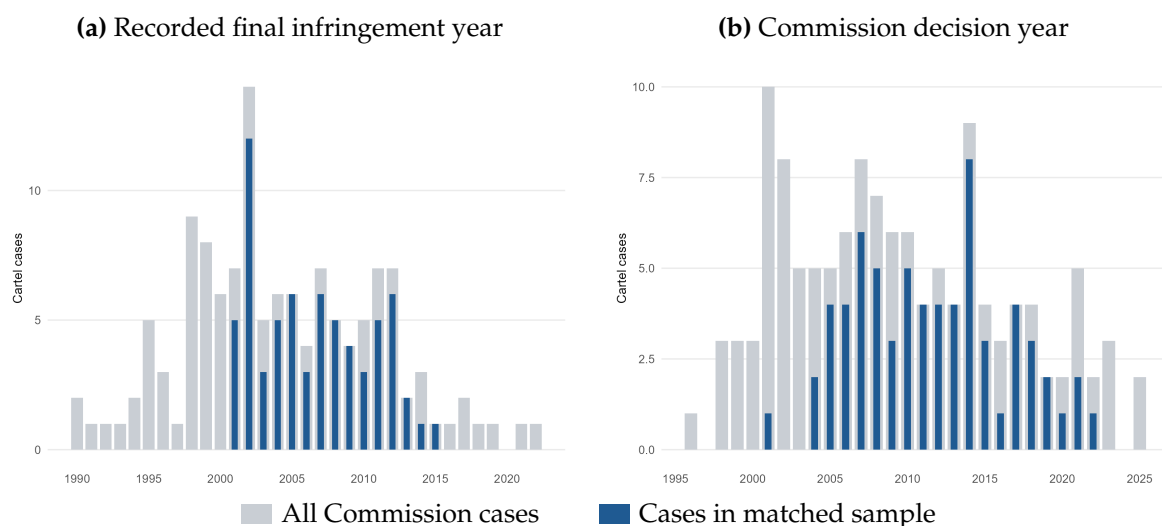


Figure 1. Cartel endings and Commission decisions

Notes. Gray bars show all 129 cases in the Commission cartel database. Blue bars identify the 67 cases represented in the analysis. Panel (a) organizes cases by the recorded final infringement year, which defines event time. Panel (b) organizes cases by the year in which the Commission published its decision.

rather than an outdated enforcement record. Decisions in the database extend through 2025, and some proven infringements end as late as 2022, but the later cases either do not yet provide the full event window or lack a member with the required financial history and comparison support (Appendix Table A.1). Figure 1 shows both dates and illustrates a basic feature of the setting. The recorded end of the conduct and the public enforcement decision are separated in time, often by several years.

The represented cartels are broad in scope. Their members are headquartered in 25 countries and operate in 84 four-digit NACE industries; 43 cases cover the EU or EEA, 17 a selected group of countries, and five are described as worldwide, with a median proven duration of 5.75 years. The longest recorded infringement in the matched sample, an animal feed phosphates cartel, lasted 35 years. The matched cases have the same median proven duration and a very similar mix of conduct as the full case database, although they tend to involve somewhat more firms. Appendix Figure A.1 shows the industry and country composition, and the cartel catalog in Appendix Table A.9 lists the affected activity, geography, conduct period, decision year, and represented members for every case.

Bureau van Dijk supplies company category, headquarters country, NACE codes, current assets, long-term debt, and employment. The financial variables are measured in $g - 1$, before the recorded ending, and enter the comparison procedure in logarithms. Because these requirements select established cartel members with observable accounts, the estimates should be read as applying to documented members with adequate firm data rather than to every legal entity mentioned in a Commission decision.

4.2 Minority-stake transactions

The Zephyr transaction data cover completed deals from 1997 through 2025, reporting the completion year, the type of transaction, the parties, and the final ownership share when available.¹⁰ We retain transactions classified as minority investments and remove deals whose known final stake is at least 50 percent; if the percentage is not reported, we use the provider’s minority-deal classification. The observed stakes are economically varied. A numerical final share is available for 17 of the 39 qualifying post-ending deals, with a median of 15.8 percent, and the remaining transactions enter through the provider’s classification.¹¹

Transactions can involve several acquirers or several targets. We first expand each record into the concrete acquirer–target pairs that it contains and then assign horizontal status separately to each pair, so that one overlapping pair in a complex transaction cannot turn every buyer and target in that transaction into a horizontal match.

A pair is horizontal when the acquirer and target share a four-digit industry code in at least one of four systems: SIC, NACE, NAICS, or Bureau van Dijk’s own classification.¹² The main definition uses primary activities together with secondary activities that can be traced to the individual firm. We include traceable secondary codes deliberately. For diversified firms, the activities in which two companies actually compete are often recorded as secondary lines, and primary codes alone would miss them. The use of several systems reduces dependence on any one coding convention. Both choices also widen the definition mechanically—a firm with many recorded codes has more opportunities to overlap—so Section 6.4 reports the estimates under primary-only coding and under measures that require shared industries to make up a minimum fraction of the two firms’ combined code lists.

Four-digit overlap measures business relatedness, not a legal antitrust market.¹³ We separately record whether a target operates in the NACE industry assigned to the cartel event.

For each firm and year, the acquirer outcome equals one if the firm completes at least one horizontal minority acquisition; the target outcome equals one if it receives at least one. A firm with several qualifying deals in one year still contributes one. This extensive-margin definition is natural for a rare event and gives each coefficient a direct interpretation as a change in annual acquisition probability. We also construct indicators for non-horizontal and for all minority investments, and a signed direction measure that equals one in a year with

¹⁰The full Zephyr extract records roughly three million corporate transactions worldwide, about 1.3 million of which are classified as minority stakes (as of October 2025).

¹¹This incomplete coverage prevents stake size from becoming the main outcome, and it means the outcome captures reported minority-stake acquisitions rather than certified passive holdings. Appendix Table A.4 reports the transaction counts and the available ownership information.

¹²The codes are those reported on the Zephyr deal record; secondary activities enter only when they can be traced to the specific firm through its own deal records and the firm panel, rather than being merged from a single current company code list.

¹³The distinction matters for diversified firms. A cartel member may be treated because of its role in one product line but acquire a target that overlaps with another part of its business. All references to the “assigned” or “cartel-associated” industry use the four-digit NACE code attached to the case, with a broader three-digit check in Section 6.4.

Table 1. Matched sample and annual minority-investment rates

	Value	
<i>Panel A. Sample</i>		
Treated firm–case events		246
Unique named cartel members		235
Unique comparison firms		886
Cartel cases		67
Ending years / countries / four-digit NACE industries	15 / 25 / 84	
Event-year range	2001–2015	
Commission-decision range	2001–2022	
Median proven duration across cases, years	5.75	
Median years from infringement ending to Commission decision	5	
Median firms reported per cartel	11	
EU/EEA-wide / selected-country / worldwide / unclassified	43 / 17 / 5 / 2	
Endings before documented enforcement / enforcement-linked	37 / 30	
<i>Panel B. Annual probability in percent</i>		
	Treated	Weighted controls
Acquirer, $t = -4, \dots, -1$	1.83	1.71
Acquirer, $t = 1, 2, 3$	3.93	1.55

Notes. Panel B averages the annual indicator across the stated event years. Comparison-firm rates use the matching weights. These raw comparisons are descriptive and need not equal the event-study estimates.

horizontal acquisitions only, minus one in a year with non-horizontal acquisitions only, and zero otherwise—including both when no investment occurs and when both types occur in the same year. This measure summarizes the direction of a firm’s annual investment activity.

4.3 The matched sample before estimation

Table 1 describes the sample and the raw acquisition rates. The 235 cartel members are compared with 886 distinct firms that are never named as cartel members in our data; a comparison firm can support more than one treated event, so the number of distinct comparison firms is smaller than the number of weighted firm-event comparisons.

Horizontal acquisitions are rare before the event. Members act as acquirers in 1.8 percent of firm-years from $g - 4$ through $g - 1$, close to the 1.7 percent rate among their weighted comparisons. In the first three full years after the ending, the raw rates are 3.9 percent for members and 1.6 percent for comparison firms. These raw comparisons pool different calendar years and do not yet use the staggered event-study design.

5 Empirical Design

The empirical challenge is to construct the acquisition path a cartel member would have followed had the recorded infringement not ended when it did. Cartel members are not a

Table 2. Balance between cartel members and comparison firms

Covariate	Member mean	Weighted comparison mean	SMD
Any horizontal minority acquisition in the four pre-event years	0.0407	0.0407	0.000
Log current assets	19.176	18.957	0.097
Log long-term debt	12.656	12.817	−0.020
Log employment	7.188	7.154	0.015

Notes. Financial variables are measured in $g - 1$. SMD is the standardized difference between the member mean and the weighted mean among comparison firms.

random set of firms. They are often large, diversified, and active in acquisitions, and their cartels end in different years. We address both problems in two steps: a separate set of comparison firms for each member and cartel event, combined in a staggered event study.

5.1 Constructing the comparison group

For every member event, the comparison pool contains firms observed in the same year that are never named in our cartel concordance. Candidates must have the same company category and operate in the same three-digit NACE industry as the member, but in a different four-digit NACE industry. The first restriction keeps broad business conditions similar; the second reduces the chance that a comparison firm is itself directly exposed to the ending of the same narrow cartel. This industry rule involves a deliberate trade-off. A firm in the same four-digit industry would appear more comparable, yet it may respond to the cartel ending even though it was not a member, which would absorb part of the effect into the counterfactual. A firm just outside the four-digit industry is less likely to be directly treated, though adjacent markets can still be affected. Section 6.6 therefore also estimates the response of non-member firms inside affected industries, which helps distinguish a member-specific change from a market-wide one.

Members and candidates must agree on whether they made any horizontal minority acquisition during the four years before g . Among the remaining candidates, we retain up to 20 firms closest in current assets, long-term debt, and employment in $g - 1$, and choose nonnegative weights, summing to one, that reproduce the member's financial profile as closely as the available firms allow. Neither the ending year nor any later outcome enters the selection. Table 2 shows that the weighted comparisons match members closely on all matching variables. Appendix Table A.5 shows that the estimates are similar when we cap the weight placed on any single comparison firm and draw on broader candidate pools.

When we study a different type of acquisition, the outcome used to form the match changes accordingly—for a non-horizontal outcome, candidates must agree on pre-event non-horizontal activity—so that each comparison stays relevant to the outcome being studied. The composition analysis in Section 6.3 instead holds the comparison fixed. It uses one

common group based on any minority-investment activity and estimates a signed horizontal-minus-non-horizontal outcome, so that conclusions about investment direction come from the same firms and weights.

5.2 Event-study specification

The estimation compares, for each cartel member, the change in its acquisition outcome from the last pre-ending year to a given event year with the corresponding weighted change among its comparison firms, and then averages these member-specific differences across cartel endings. The coefficients therefore measure whether cartel members change their acquisition behavior more or less than otherwise similar firms around the ending.

Formally, let Y_{ist} denote the acquisition outcome of firm i in calendar year t within the member-specific matched group s , and let g_s denote the final infringement year assigned to that group. The member indicator M_{is} equals one for the named cartel member and zero for its comparison firms. A convenient representation of the comparison is

$$Y_{ist} = \alpha_{is} + \lambda_{st} + \sum_{\substack{\ell=-4 \\ \ell \neq -1}}^4 \beta_{\ell} (M_{is} \times \mathbf{1}\{t - g_s = \ell\}) + \varepsilon_{ist}, \quad (1)$$

where α_{is} absorbs permanent differences between a member and its comparison firms, λ_{st} absorbs shocks common to a matched group in a calendar year, and $\ell = -1$ is the reference period. Because cartels end in different years and effects may vary across ending cohorts, we do not estimate equation (1) as a conventional two-way fixed-effects regression. We instead estimate the dynamic effects with the heterogeneity-robust procedure of de Chaisemartin and D’Haultfœuille (2024), which forms the change for each treated member relative to its matched firms and aggregates across ending cohorts; firms whose own cartel has already ended never serve as controls for later cohorts.¹⁴

We report three pre-event coefficients, the ending year, and four subsequent years. The final year of the infringement is $\ell = 0$, and $\ell = 1$ is the first complete post-ending year. The primary summary is the average of β_1 , β_2 , and β_3 , the first three complete years after the documented ending, over which the increase is sustained.

5.3 Identifying assumption and inference

The identifying assumption is that, absent the event, the acquisition probability of a named member would have changed like that of its weighted comparison firms. Matching makes the assumption more plausible by aligning pre-ending acquisition incidence, broad industry conditions, company type, and financial capacity. The pre-event coefficients provide a further check on whether the annual acquisition paths remain aligned before the ending. What neither matching nor a flat pre-event path can rule out is a shock that arrives at the same time

¹⁴Ordinary two-way fixed effects can combine such staggered comparisons with unintuitive weights; see de Chaisemartin and D’Haultfœuille (2020), Callaway and Sant’Anna (2021), and Sun and Abraham (2021).

as the ending and affects cartel members differently, above all the enforcement episode itself. This is the principal remaining threat, and the results address its most plausible forms directly: false event dates within the cartel period, the later Commission decision, the classification of endings by enforcement timing, transaction direction, target location, and the response of non-members in the affected industry.

The event-study intervals in the figures cluster observations by firm, because the same company can appear in more than one matched group.¹⁵ Members of the same cartel also face a common ending, so for the three-year average we additionally allow outcomes to be correlated within cartel cases. Because the 67 cartels differ greatly in size, we accompany the cartel-clustered interval with a leave-one-cartel-out jackknife and a restricted wild-cluster bootstrap, the combination recommended for unequally sized clusters (MacKinnon et al., 2023). Table 4 in Section 6 reports the results; the point estimate is identical throughout, and only the attached uncertainty changes.

6 Results

6.1 The change in the raw data

We begin with the acquisition paths themselves. Figure 2 plots the annual rate at which cartel members and their weighted comparison firms acquire horizontal minority stakes. The figure shows raw rates, with no estimation behind them.

Before the ending, the two series move together at low levels. Members acquire a horizontal minority stake in 1.8 percent of firm-years, compared with 1.7 percent among their weighted comparisons. The paths separate in the first full post-ending year. Over years one through three, the member rate rises to 3.9 percent while the comparison rate is 1.6 percent, so the raw gap widens by 2.3 percentage points. This simple difference ignores the staggered timing of the endings, but it shows that the main result is already present in the underlying rates.

6.2 The dynamic response

Figure 3 follows cartel members from four years before through four years after the final year of the proven infringement. Panel (a) asks whether they acquire horizontal minority stakes; panel (b) asks whether they become the targets of such investments. Every point compares the member's change with the weighted change among its matched firms, relative to the last pre-ending year.

The buyer-side path changes only after the cartel ends. The three pre-event coefficients are close to one another and jointly indistinguishable from zero, and acquisition activity changes little during the ending year itself. It then rises in the first full year, remains at almost the

¹⁵"Firm" here means the underlying company: a firm that appears in several matched groups, whether as a member or as a comparison, forms a single cluster.

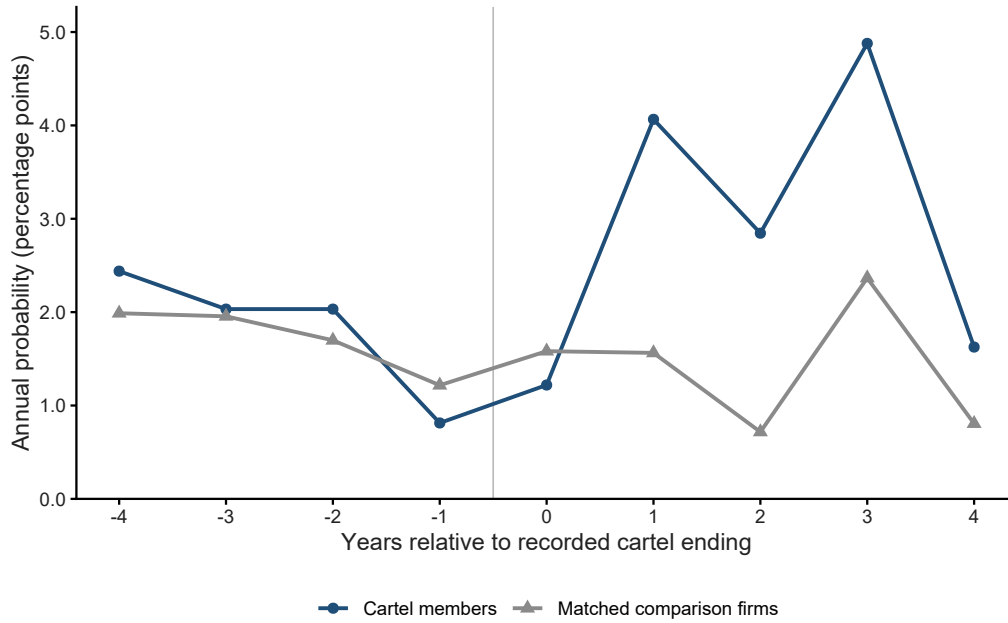


Figure 2. Horizontal minority acquisitions before and after cartel ending

Notes. The lines show annual acquisition rates for named cartel members and their weighted comparison firms at each event time. The vertical line marks the boundary between the pre-ending and post-ending periods; the recorded final infringement year is $t = 0$. The figure reports levels and applies no event-study normalization.

Table 3. Average effects during the first three post-ending years

Outcome	Member as acquirer	Member as target
Horizontal minority stake	2.78 [0.96, 4.61]	-0.15 [-2.08, 1.78]
Horizontal minus non-horizontal direction	4.29 [1.85, 6.73]	1.65 [-2.50, 5.81]
Any minority stake	0.17 [-2.13, 2.46]	-0.70 [-4.33, 2.93]

Notes. Entries average $t = 1, 2, 3$, with firm-clustered 95 percent confidence intervals in brackets. The first and third rows are percentage-point changes in annual probability; the second row is the change in the signed direction index of Section 4.2, scaled to percentage points. The first row uses the horizontal-acquisition comparison. The last two rows use one common comparison group based on any pre-event minority-investment activity.

same level through year three, and falls back by year four, when the estimate is no longer statistically distinguishable from zero. The average increase during years one through three is 2.8 percentage points—large relative to the 1.8 percent annual rate before the ending.

The transaction counts reinforce the extensive-margin interpretation. After the ending, the number of members making horizontal minority acquisitions more than doubles, while aggregate deal volume changes little. The estimate reflects a widening of a rare behavior across firms, which is also why the extensive-margin indicator is the natural outcome.

The target side, by contrast, shows little movement. Its estimates stay near zero before and after the event, and the average post-ending change is slightly negative. Buyers and targets need not move symmetrically, since treated members can acquire stakes in firms that never participated in a cartel and hence never appear on the treated target side.

Table 3 summarizes the average effects.

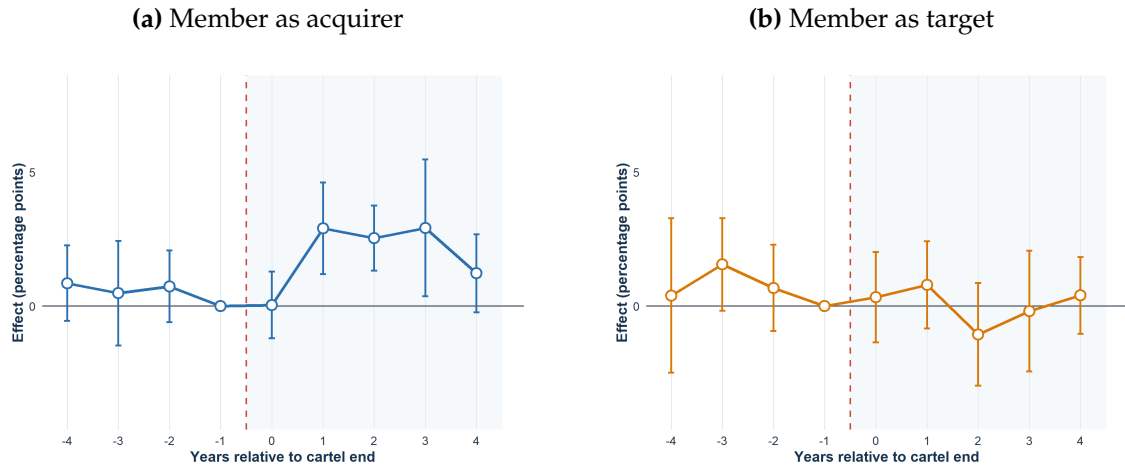


Figure 3. Horizontal minority-stake activity around cartel ending

Notes. The figure reports dynamic difference-in-differences estimates for named cartel members relative to matched firms that are never named in a Commission cartel case. Outcomes are firm-year indicators. The recorded final infringement year is $t = 0$, and $t = -1$ is the reference period. Bars are 95 percent confidence intervals clustered by firm.

Table 4. Inference for the average acquirer effect

Inference	Clusters	95% CI	p -value
Firm-clustered	1,121	[0.96, 4.61]	0.003
Cartel-clustered	67	[0.46, 5.11]	0.020
Clustered by treated four-digit NACE industry	84	[0.18, 5.39]	0.036
Leave-one-cartel-out jackknife	67	[0.44, 5.13]	0.021
Wild-cluster bootstrap	67	[0.52, 5.05]	0.014

Notes. The point estimate is 2.78 percentage points in every row: treatment, matching, and the average acquirer estimate over $t = 1, 2, 3$ are identical, and only the measurement of uncertainty changes. The first row clusters by firm. The second and third rows cluster by the 67 assigned cartel cases and by the member's four-digit NACE industry, with t distributions at 66 and 83 degrees of freedom. The jackknife and the bootstrap operate at the cartel level: the jackknife omits one cartel at a time, and the wild-cluster bootstrap uses Webb weights and 99,999 draws.

The figures cluster by firm. The economic event, however, occurs at the cartel level, so Table 4 shows that the result is similar when statistical uncertainty is allowed to be correlated at the cartel or industry level, and under a leave-one-cartel-out jackknife and a wild-cluster bootstrap. The estimate remains statistically significant at the five percent level throughout.

The decline in the annual coefficient after year three concerns the flow of new deals, not the links already formed. To translate the annual extensive-margin response into deal flows, Figure 4 sums the estimated yearly changes in the *number* of new horizontal minority acquisitions, an outcome that, unlike the annual indicator, counts multiple deals by the same firm. By the end of year three the summed response reaches 15.2 additional acquisitions per 100 member events, and it grows further in year four. The two scales are consistent. The annual indicator effects sum to roughly eight percentage points over the three years, and the count measure exceeds this because several members complete more than one deal.¹⁶

¹⁶We call this a cumulative acquisition response rather than an ownership stock because the transaction data

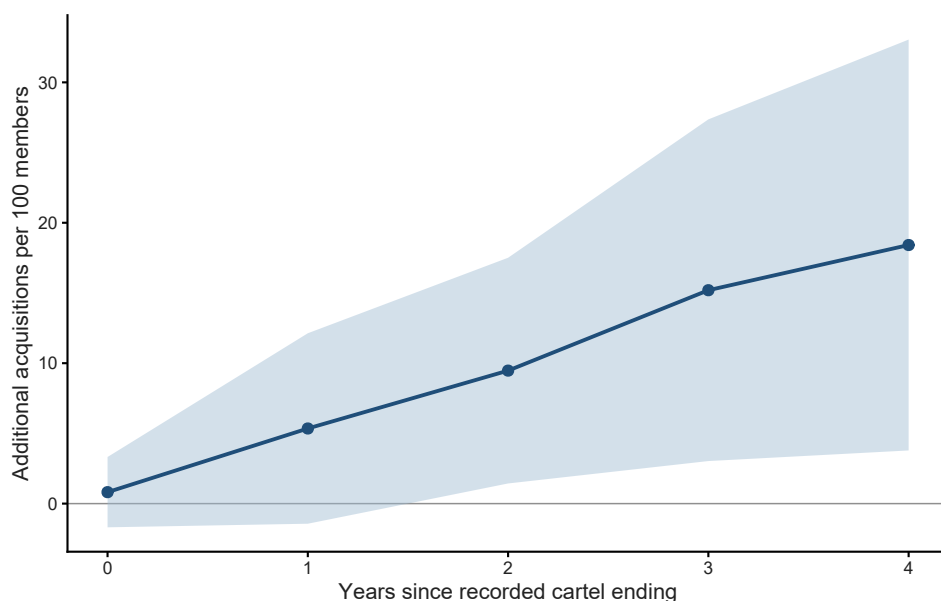


Figure 4. Cumulative new horizontal minority acquisitions

Notes. The line cumulates the estimated effect on the annual number of new horizontal minority acquisitions from the recorded ending through each event year. Values are acquisitions per 100 member events; unlike the annual indicator, this outcome counts several deals by the same firm. The underlying annual estimates are 0.82 in the ending year and 4.53, 4.12, 5.73, and 3.22 in years one through four; the plotted series sums them from $t = 0$ onward, reaching 5.35, 9.47, 15.20, and 18.42 by the ends of years one through four. Bars are 95 percent confidence intervals clustered by assigned cartel case.

6.3 More investment, or a different direction?

The increase in horizontal acquisitions could be part of a general wave of corporate transactions. Figure 5 compares the average response across four outcomes, each with comparison firms selected on pre-event activity in that outcome. Horizontal minority acquisitions rise by 2.7 percentage points, while all minority investments, horizontal full acquisitions, and all full acquisitions show no comparable increase.

We next ask whether members redirect their minority investments toward horizontally related firms, using the same members, comparison firms, and weights for horizontal, non-horizontal, and total minority activity. The outcome in Figure 6 records the direction of a firm's annual activity: it equals one in a year with horizontal acquisitions only, minus one in a year with non-horizontal acquisitions only, and zero otherwise—both when nothing is acquired and when both types occur. An increase means that the mix of a firm's minority-investment activity has moved toward horizontal targets. Because inactive years enter as zeros, the index does not measure the choice of target conditional on investing; it summarizes how often each type of year occurs.

With the same firms and weights behind every row, Table 5 separates the change into its components. The horizontal component rises by 2.3 percentage points, the non-horizontal component falls by about two points, and the probability of making *any* minority investment barely changes. Horizontal-only investment years become more frequent after the ending

do not record every subsequent divestment.

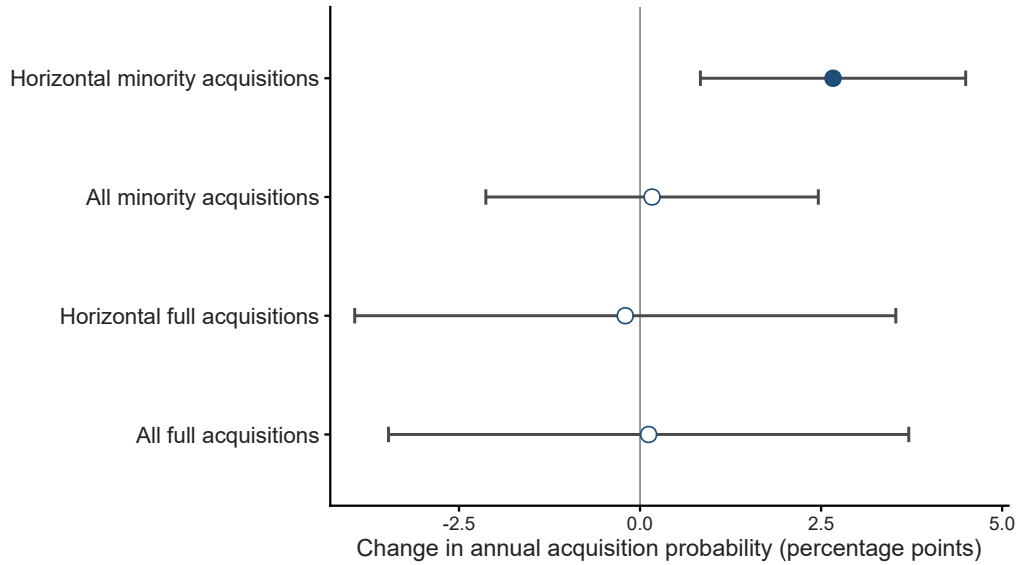


Figure 5. Minority stakes and acquisitions of control

Notes. Points are average acquirer effects over $t = 1, 2, 3$. Each outcome uses comparison firms selected on the same predetermined firm characteristics and pre-event activity in that outcome. Horizontal status is defined for the concrete acquirer–target pair. Bars are 95 percent confidence intervals clustered by firm.

Table 5. The composition of minority investment

Outcome	Effect	95% CI
Horizontal minority stake	2.31	[0.50, 4.11]
Non-horizontal minority stake	−1.98	[−3.70, −0.26]
Any minority stake	0.17	[−2.13, 2.46]
Horizontal minus non-horizontal direction index	4.29	[1.85, 6.73]

Notes. Acquirer-side effects averaged over $t = 1, 2, 3$, in percentage points. All rows use the same member events, comparison firms, and weights, selected on any pre-event minority-investment activity. The last row is the signed direction index of Section 4.2, scaled to percentage points. Confidence intervals cluster by firm. Appendix Table A.6 reports pre-event diagnostics and the destination split.

and non-horizontal-only years less so. The direction index summarizes the two movements. Stable before the ending, it turns positive afterward, with an average increase of 4.3 percentage points. In other words, members redirect part of their minority investment from non-horizontal toward horizontal targets. The target-side index is positive but imprecise. Appendix Figure A.2 repeats the analysis with outcome-specific comparison groups, and Appendix Table A.6 reports the destination split.

6.4 Where the new stakes go

Section 3 left open whether an ownership response should target the affected market or reach across the acquirer’s businesses. None of the 39 acquired firms in years one through three was a member of the same cartel. Twelve targets operate in the four-digit NACE industry assigned to the cartel event; the other 27 overlap with a different part of the acquirer’s business. Replacing the four-digit boundary with the broader three-digit industry moves

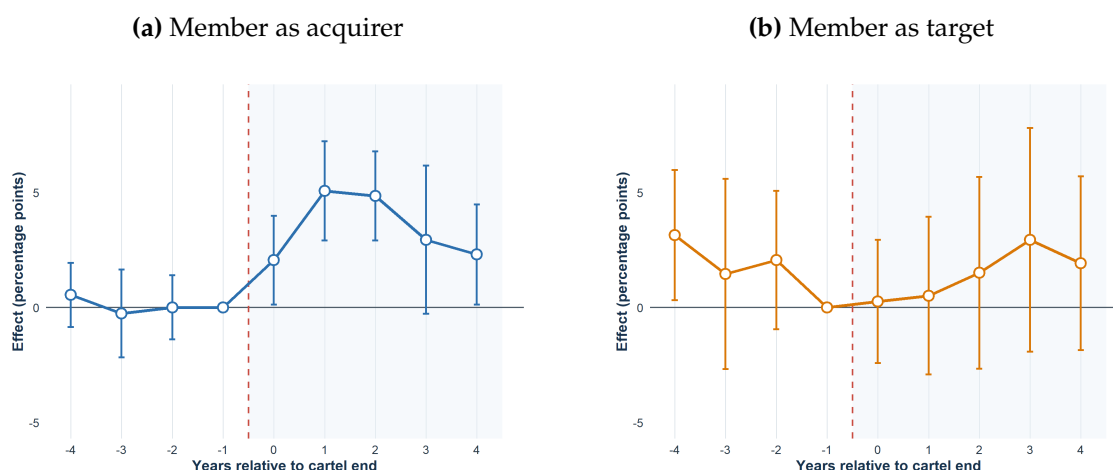


Figure 6. The direction of minority investment

Notes. The outcome equals one in a year with horizontal acquisitions only, minus one in a year with non-horizontal acquisitions only, and zero when both or neither occur. Every component within a panel uses the same member events, comparison firms, and weights. The reference period is $t = -1$. Bars are 95 percent confidence intervals clustered by firm.

only two deals into the assigned-market category. Event studies that split acquisitions by destination (Appendix Figure A.3) show increases in both categories, with the larger share outside the assigned industry and no statistically significant difference between the two. Appendix Table A.7 lists all 39 transactions individually.

Two publicly documented examples illustrate the two destinations. Within a year of the recorded end of the E.ON–GDF market-sharing infringement in 2005, E.ON assembled minority positions in Central European gas and power utilities, among them a non-controlling stake of just over 49 percent in Pražská plynárenská, the Prague gas distributor. These were new financial links in the very activity the agreement had covered.¹⁷ Robert Bosch, whose participation in the braking-systems and spark-plugs conduct ended in 2011, instead joined an EUR 18 million financing round of Heliatek, a Dresden producer of organic solar films, in 2014, a target far from vehicle parts but overlapping with Bosch’s then-active photovoltaics business.¹⁸ The two cases mark the range of the catalog in Appendix Table A.7, from a large direct stake inside the affected activity to a small position in a co-investment round far outside it.

The destination pattern is broader than the narrow substitution account. Former cartelists do not buy their previous partners; they build horizontal links across their portfolios, of which the cartel market is only one part. One reading is that the end of a cartel prompts a firm-wide review of competitive relationships. Another is that four-digit codes measure the relevant market too coarsely, so that some deals recorded as “outside” are closer to the cartel

¹⁷European Commission, merger case COMP/M.4238 (decision of 11 July 2006), which records E.ON’s non-controlling 49.35 percent shareholding in the distributor alongside a jointly controlling stake in its holding company acquired in a share swap with RWE.

¹⁸Heliatek, “Heliatek completes 18 million euro in Series C funding round,” press release, September 2014; the round was led by AQTON SE with participation of Bosch, BASF, Wellington Partners, innogy Venture Capital, eCAPITAL, High-Tech Gründerfonds, and TGFS.

product than the classification suggests. Either way, the pattern is broader than the literal replacement of a former cartel partner, and Section 7 returns to its interpretation.

Because the horizontal classification is deliberately broad, with four coding systems and secondary activities included, a diversified firm has many opportunities to share at least one code with a target. The increase remains positive under primary-only coding and under increasingly restrictive measures of industry overlap (Appendix Table A.5 and Figure A.4).

A random-target benchmark tests the definition directly. For each minority deal the members complete during years one through three, we ask how often a randomly drawn target would classify as horizontal to the same acquirer, drawing pseudo-targets from minority transactions completed in the same year and country and with a similarly rich industry-code profile. Under the main definition, 37.9 percent of the members' 103 actual deals are horizontal, against a random benchmark of 21.6 percent (95 percent range 14.6 to 29.1); none of 9,999 random assignments reaches the observed share (empirical p -value < 0.001). Under primary-only coding the gap persists: 22.3 percent of actual deals are horizontal, against a benchmark of 12.4 percent ($p = 0.003$). The horizontal tilt of post-ending investment reflects the targets the members choose, not the length of diversified buyers' code lists.

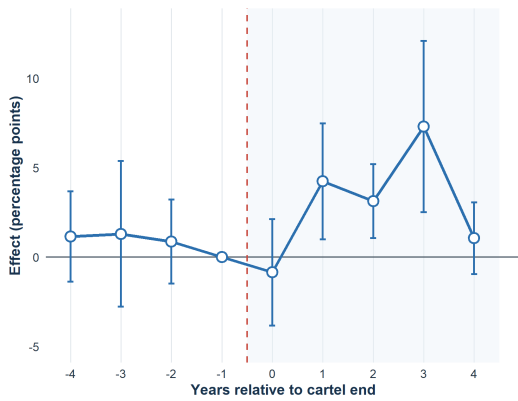
6.5 Timing and alternative explanations

The post-ending timing is distinctive. Moving the event two, three, or four years into the cartel period, and using only observations that still precede the true ending, produces no comparable rise in acquisition activity (Appendix Figure A.6). The Commission decision offers a different benchmark. Decisions arrive a median of five years after the proven conduct ends, and when event time is organized around the decision year, acquisition activity has already started to move (Appendix Figure A.7). The formal decision and fine are too late to explain the beginning of the response. An enforcement interpretation is not eliminated by this timing alone, since an immunity application or inspection can lie close to the final infringement year even when the published decision does not.

We address that possibility with the classification from Section 2.1. Panel (a) of Figure 7 restricts the analysis to the 37 cartels whose proven conduct ended before the first documented enforcement step. The pre-period is quiet and the average post-ending effect is 4.9 percentage points [1.5, 8.3]. Panel (b) reports the 30 cartels in which a leniency application, complaint, inspection, or other authority action preceded or coincided with the ending; their average effect is 1.4 points [−0.5, 3.4]. The pattern is what one would expect if the response reflects the end of coordination rather than enforcement itself. The effect is concentrated where no enforcement step precedes the ending, though the difference between the two subsamples is estimated with some uncertainty. Panel (a) also shows directly that the main result is not generated by cases in which the recorded ending and visible enforcement coincide.

Permutation exercises ask whether a change of this size could arise by chance (Figure 8). The first assigns the cartel ending to a randomly chosen comparison firm within each matched

(a) Ending before documented enforcement



(b) Enforcement linked to the ending

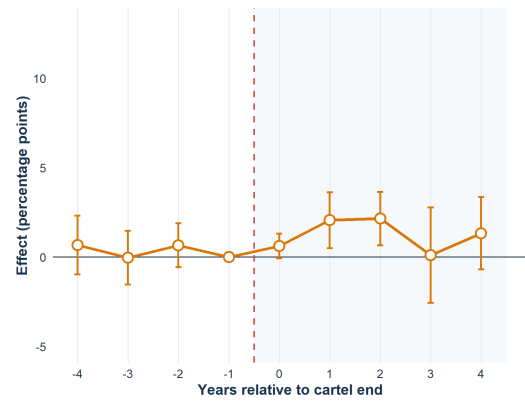


Figure 7. Acquirer response and the timing of enforcement

Notes. Panel (a) contains 37 cartels and 96 matched member events for which the last proven infringement date precedes the first documented leniency application, complaint, inspection, or other authority action. Panel (b) contains 30 cartels and 150 matched events in which enforcement preceded or coincided with the ending. Bars are 95 percent confidence intervals clustered by firm. The reference period is $t = -1$.

group, while excluding the actual member. Across 999 draws, none produces an estimate as large as the one observed for cartel members ($p = 0.001$). The second exercise keeps the members and their ending dates in place, but randomly reverses each cartel's contribution to the estimate. The observed increase again lies in the far tail of the distribution (two-sided $p = 0.009$).

The comparison design itself matters little for the broad conclusion. The estimate remains positive under caps on the weight of any single comparison firm, wider candidate pools, one event per firm, equal total weight per cartel case (5.5 points, cartel-clustered $p = 0.017$), and the earlier matching window that reserves $g - 4$ through $g - 2$ for validation; removing each cartel, member country, and ending cohort in turn also leaves every estimate positive. Appendix Tables A.5 and Figures A.5 and A.8 report these exercises.

6.6 The scope of the response

The response does not belong to the entire affected industry. Samples that assign the cartel ending to all firms in the associated four-digit NACE industry produce small estimates near zero (Appendix Table A.8).

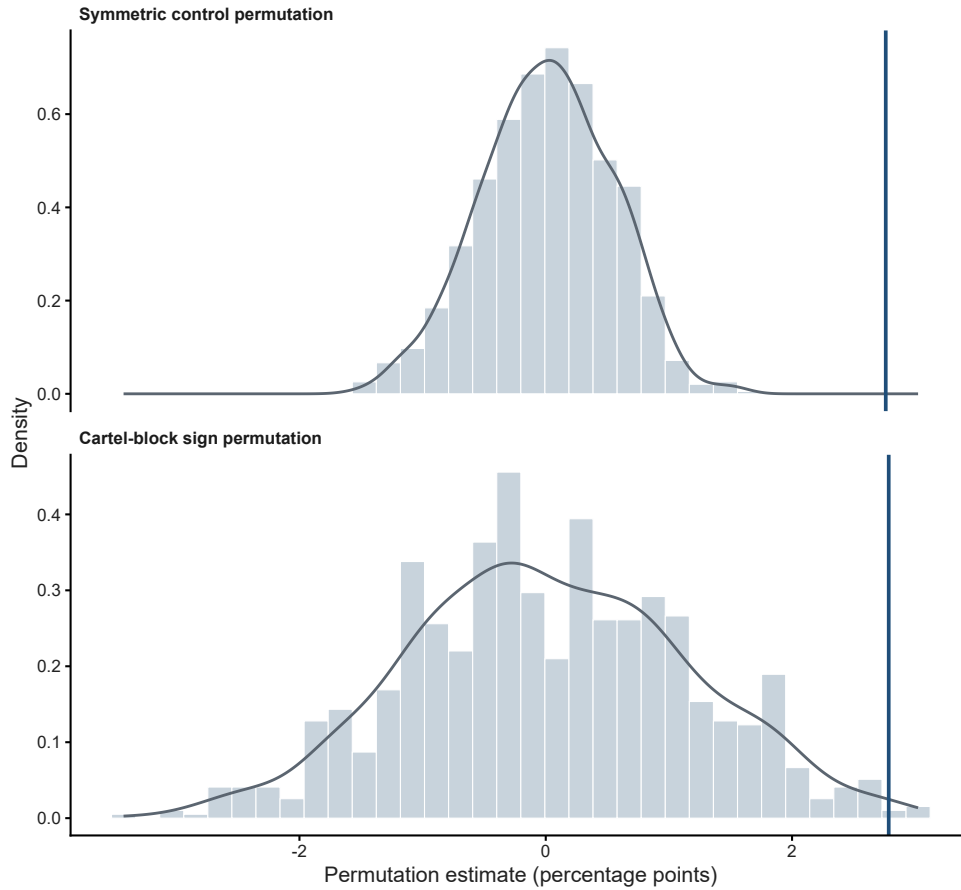


Figure 8. Permutation inference for the average acquirer effect

Notes. The top panel assigns, within each matched group with at least two never-named comparison candidates (224 of 246 events), one candidate the cartel ending with equal probability and compares it with the remaining candidates; actual members are excluded. The bottom panel keeps all 246 member events and randomly flips the sign of all member-event contributions within each of the 67 cartels. Each panel contains 999 draws, centered on zero by construction. The vertical line marks the actual average effect over $t = 1, 2, 3$. Empirical p -values: 0.001 (top) and 0.009 (bottom, two-sided).

7 Discussion

The change appears only after the ending, and it belongs to the cartel members themselves rather than to the average firm in the affected industry. It is concentrated in minority stakes, and it shifts the direction of investment toward horizontal targets while total deal activity stays flat. Together, these facts describe a reorganization of competitive relationships, not a generic merger cycle.

Partial-ownership theory provides a natural interpretation. During collusion, the agreement itself limits the actions through which one firm takes business from another; once it ends, a financial interest in another firm again changes the return to competing aggressively. A minority stake can recreate part of the alignment that explicit coordination had provided while leaving both firms in the market, and the delay between the infringement ending and the completed transactions fits the time needed to identify a target and negotiate a stake. The transaction destinations show that this reorganization is broader than the narrowest version

of the mechanism. Former cartelists do not buy their previous partners, and most targets lie outside the four-digit industry attached to the case. Members appear to respond to the end of one cartel by building horizontal links across several businesses, not by reconstructing the specific relationship the Commission dissolved.

Could the response instead reflect the enforcement episode rather than the loss of coordination—a strategic review after a raid or fine, working through management turnover, compliance reform, or financing? Several facts are difficult to reconcile with that reading as the whole story. The response is horizontal rather than general, absent from full acquisitions, present at false dates nowhere inside the cartel period, and already underway before the eventual Commission decision. Most tellingly, it appears—with a larger point estimate—among the cartels whose conduct ended before any documented enforcement step, where the recorded ending cannot be an echo of visible enforcement, although the difference between the enforcement subsamples is itself imprecise. An enforcement shock may well contribute; it does not easily generate this particular combination of patterns.

The result joins a set of findings on what changes when explicit collusion stops. Prices fall; innovation, which rises during collusion, reverts to pre-collusion levels (Kang, 2025); consolidation through mergers can follow (Davies et al., 2015; Dong et al., 2019); and firms adjust how they communicate publicly (Bourveau et al., 2020). Heim et al. (2022) show the complementary margin *ex ante*. Minority acquisitions rise when leniency programs make explicit collusion more costly. Together, these studies suggest that explicit coordination is one instrument among several for softening rivalry, and that its withdrawal moves firms along the remaining margins, among them partial ownership.

The transactions at issue are also the ones competition authorities are least likely to see. When one competitor acquires a minority stake in another, concentration measures do not move and control-based merger review is typically not triggered, yet the investment transfers cash-flow rights and sometimes voting power, board access, or information. The parallel to below-threshold mergers is direct: Wollmann (2019) shows that transactions exempt from premerger notification proceed essentially unexamined, and non-controlling stakes occupy the same blind spot for a different legal reason. United States merger review already treats a collusive past as a warning sign. The 2023 Merger Guidelines describe evidence that firms “previously engaged in express or tacit coordination” as “highly informative” about a market’s susceptibility to coordination (U.S. Department of Justice and Federal Trade Commission, 2023). The principle, however, operates only on transactions the agencies review. The Delivery Hero–Glovo decision demonstrates that such links can matter for conduct. Our evidence identifies cartel endings as a moment when horizontal ownership links become systematically more likely—a natural screening trigger for authorities that already know, from their own decisions, exactly which firms are involved.

Two limits remain: the recorded ending can coincide with an investigation or a broader change in corporate strategy, and the data do not reveal the governance rights attached to each stake. They do not alter the documented empirical pattern, but they limit its causal

interpretation. After the documented end of a cartel, former members become more likely to acquire minority stakes in horizontally related firms.

8 Conclusion

Cartels are usually studied while they operate. Much less is known about how their members reorganize once explicit coordination ends. This paper follows firms found by the European Commission to have participated in cartels and studies the ownership links they form around the final year of the proven infringement.

The main change appears on the buyer side. During the first three full years after the ending, members become substantially more likely to acquire horizontal minority stakes, an increase spread across many members rather than concentrated among a few. We detect no corresponding change on the target side, full acquisitions do not display the same increase, and the same firms' minority investments turn from non-horizontal toward horizontal targets. The change begins only after the documented ending and is not driven by any single cartel, country, or cohort.

The transactions are not a literal reconstruction of the old agreement. None of the targets is a former member of the same cartel, and most operate outside the narrow industry associated with the case. The evidence points instead to a broader reorganization of horizontal ownership by former cartelists, a systematic change in whom firms choose to invest in.

These findings widen the usual account of post-cartel adjustment. Firms can consolidate through full acquisitions, but they can also form partial ownership links while remaining separate competitors, a response that firm counts and standard market-share concentration measures do not register and that control-based merger review largely does not reach. Ending a cartel can change the network of ownership even when every company involved remains a separate firm.

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A Data Construction and Additional Results

A.1 Cartel sample coverage and composition

Figure A.1 reports the headquarters and two-digit NACE composition of the 246 matched firm–case events. The industry panel uses the member’s event-specific four-digit NACE code and pools divisions outside the nine largest categories. Table A.1 documents why the matched ending cohorts stop in 2015, and Table A.2 traces the construction of the estimation sample and the selection into it.

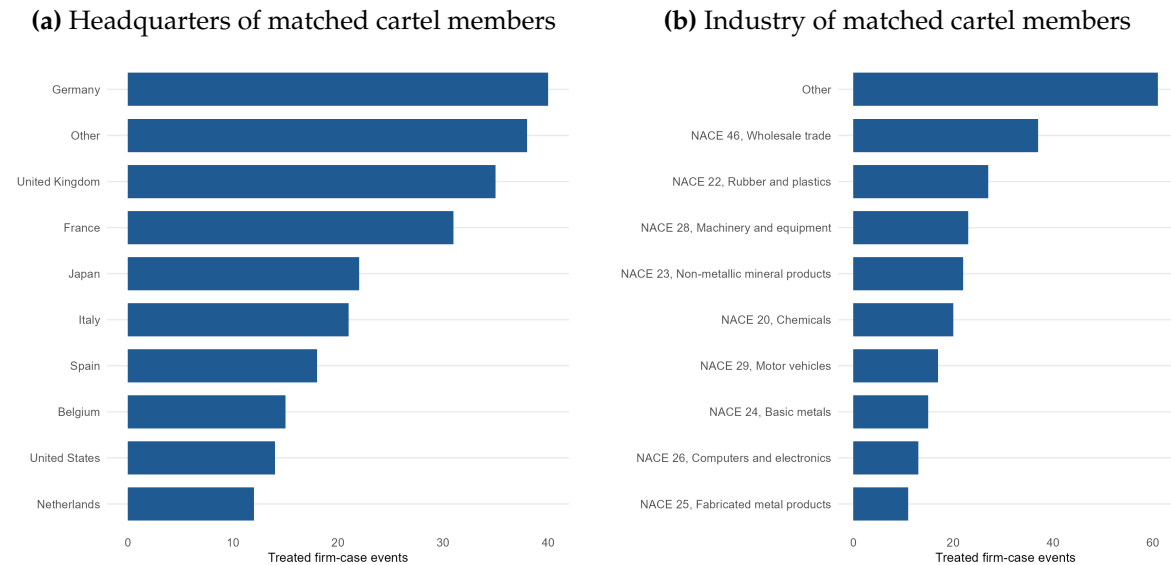


Figure A.1. Geographic and industry composition of the matched sample

Notes. Bars count treated firm–case events, not unique cartel cases. A firm can contribute to more than one cartel case. Countries and two-digit NACE divisions outside the nine largest groups are pooled.

Table A.1. Why the matched ending cohorts stop in 2015

Item	Count / year
Cartel cases in the updated database	129
Latest Commission decision year	2025
Latest recorded final infringement year	2022
Decisions during 2016–2025	27
Cases with final infringement year 2016 or later	7
Post-2015 cases entering the financially complete event pool	5
Financially complete firm events during 2016–2019	41
Those events with an eligible industry comparison	0
Those events with a comparison under the broader industry rule	0
Latest event year in either matched sample	2015

Notes. Treatment timing is the recorded final infringement year, not the Commission decision year. The broader rule permits comparison within the same two-digit NACE industry when no suitable firm exists at three digits. The underlying cartel data include Commission decisions through 2025.

Table A.2 combines the sample construction with an analysis of who is lost along the way. Panel A records the sequence from named-member events in the source panel to eligible and

matched events. Panel B compares eligible events with and without a suitable comparison firm. The two groups are about equally likely to involve a member that is absorbed in a full acquisition within five years of the ending ($p = 0.297$), so the event-window requirement does not appear to select the sample on post-ending survival. The unmatched members differ in a different way: they are larger, carry more debt, and were far more acquisition-active before the ending—firms for which no credible never-named comparison of similar scale exists. Panel C looks one step earlier and finds the same pattern: among the source-panel events that fail eligibility, 392 lack the complete nine-year window, and only eight of those involve a member recorded as a full-acquisition target after the ending.

A.2 Outcome definitions

Table A.3 summarizes the four definitions of a horizontal minority-stake transaction. Every industry system is compared at four digits. Where secondary codes are included, we use only codes that can be assigned to the concrete firm; aggregate multi-party code lists are not allocated arbitrarily across parties.

Table A.3. Four-digit horizontal-deal definitions

Industry codes	Assignment	Horizontal rule
Primary only	Deal-wide	At least one acquirer-side and one target-side party share a four-digit code. All expanded pairs inherit horizontal status.
Primary and traceable secondary	Deal-wide	The same deal-wide rule applied to all firm-specific codes.
Primary only	Pairwise	The concrete acquirer and target share a four-digit primary code.
Primary and traceable secondary	Pairwise	The concrete acquirer and target share a four-digit primary or firm-specific secondary code. Main definition.

Notes. Overlap is checked in SIC, NACE, NAICS, and Bureau van Dijk's industry classification. A match in any system is sufficient. Deal-wide rules classify a whole multi-party transaction. Pairwise rules classify each expanded acquirer–target pair.

The stricter overlap measure accounts for the number of industries attached to each firm. For classification system m , let C_i^m and C_j^m denote the four-digit industries of the acquirer and target. We calculate

$$J_{ij}^m = \frac{|C_i^m \cap C_j^m|}{|C_i^m \cup C_j^m|}. \quad (\text{A.1})$$

This ratio is high only when the firms share a substantial part of their combined industry lists. We calculate it separately in each classification system and use the largest value; the robustness exercises require that value to reach either 0.25 or 0.50.

Table A.2. Selection into the matched estimation sample

Stage	Cartel cases	Firm–case events	Unique firms
<i>Panel A. Sample construction</i>			
Commission cartel cases in the updated database	129		
Named-member event in the source firm panel	121	1,717	1,600
Financially and temporally eligible member event	82	470	438
Represented in the preferred matched sample	67	246	235
<i>Panel B. Eligible events by match availability</i>			
	Matched	Unmatched	Difference
Firm–case events	246	224	
Unique member firms	235	207	
Cartel cases	67	72	
Any pre-ending minority acquisition (percent)	8.13	38.84	-30.71***
Log total assets at $g - 1$	19.84	21.90	-2.06***
Log employment at $g - 1$	7.19	8.90	-1.71***
Log long-term debt at $g - 1$	12.66	17.22	-4.57***
Recorded cartel-ending year	2006.47	2008.34	-1.87***
Headquartered in DE, GB, FR, JP, or IT (percent)	60.57	61.61	-1.04
Cartel duration (years)	9.41	9.44	-0.03
Reported firms in the cartel case	22.28	20.06	2.21*
EU/EEA-wide cartel (percent)	60.57	70.98	-10.41**
Years from ending to Commission decision	5.06	5.28	-0.22
Full-acquisition target within five years (percent)	4.47	6.70	-2.22
<i>Panel C. Window support among source-panel events outside eligibility</i>			
	Events	Share within parent	
Not financially and temporally eligible	1,247		
Incomplete nine-year panel window	392	31.4%	
Recorded as full-acquisition target in $g, \dots, g + 4$	8	2.0%	
Complete window, other eligibility failure	855	68.6%	

Notes. Panel A uses the named-member and assigned-case indicators in the source firm panel from which the 470-event eligibility registry was constructed. Panel B entries are means across firm–case events, so cartels with many members receive more weight; the case-weighted median duration in Table 1 is therefore smaller. Eligibility requires a complete $g - 4$ through $g + 4$ window, all three matching accounts at $g - 1$, a four-digit NACE industry, and a four-year pre-event outcome history. Panel B compares eligible events retained by the matching design with eligible events for which no suitable never-named comparison firm exists. Differences are descriptive; stars report two-sided Welch tests, with *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. The five headquarters countries are the five most common in the matched sample. A full-acquisition target is identified from the compact Zephyr full-acquisition target history. The compact data distinguish a recorded full acquisition from an unexplained panel gap, but do not separately identify insolvency or legal restructuring.

Table A.4. Transactions underlying the acquirer event study

Event period	Unique deals	Unique acquirers	Cartel cases
$t = -4, \dots, -1$	46	9	11
$t = 0$	3	3	3
$t = 1, 2, 3$	39	19	18
$t = 4$	7	4	3
<i>Destination relative to the cartel-associated industry</i>			
Former co-cartelist in the assigned case	0	0	0
Target in the cartel-associated industry	12	6	6
Target in another horizontal industry	27	16	15
<i>Destination under the broader assigned NACE3 definition</i>			
Target in the assigned NACE3 industry	14	7	7
Target outside the assigned NACE3 industry	25	14	13
<i>Reported ownership shares during $t = 1, 2, 3$</i>			
Final ownership share observed		17 of 39 unique deals	
Final share at the 25th percentile, median, and 75th percentile		5.54 / 15.76 / 40.00 percent	
Initial share reported as zero / positive / missing		6 / 7 / 26	

Notes. A transaction may be attached to more than one member–case event, so 39 unique deals correspond to 42 event–deal observations during $t = 1, 2, 3$. Every unique deal receives the same NACE3 classification across its event assignments. The assigned NACE industries are descriptive proxies and not legal market definitions. Deals with a known final share of at least 50 percent are excluded. When the share is unreported, the sample relies on the provider’s minority-deal classification. Ownership-share statistics count each unique deal once. “Former co-cartelist” means a firm named in the same Commission case.

Table A.7 lists the 39 post-ending transactions individually.

A.3 Acquisition activity by transaction type

Figure A.2 repeats the analysis after selecting comparison firms separately for horizontal, non-horizontal, and total minority investment. Figure 6 in the main text is more informative about direction because it compares the three outcomes for the same firms with the same weights.

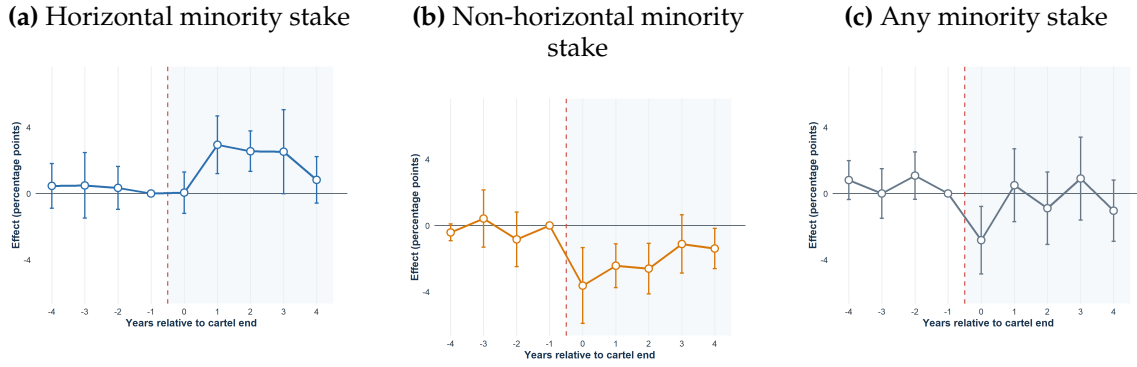


Figure A.2. Acquisition activity by transaction type

Notes. The panels report acquirer-side event studies for horizontal minority stakes, non-horizontal minority stakes, and any minority stake. Each outcome has its own comparison weights, constructed within the predetermined candidate pool using pre-event activity in that outcome. The reference period is $t = -1$. Confidence intervals cluster by firm.

A.4 Destination of the acquisitions

Figure A.3 shows the event studies behind the destination counts in Section 6.4, separating acquisitions by whether the target operates in the four-digit NACE industry assigned to the cartel event. Figure A.4 shows the average effects under the stricter relatedness measures based on equation (A.1), and Table A.5 collects the estimates across measurement rules and comparison designs. Under the all-known-industries definition, classifying each buyer–target pair separately or classifying the transaction as a whole yields the same estimate; under primary-only coding the two assignment rules differ slightly (1.42 versus 1.28 percentage points).

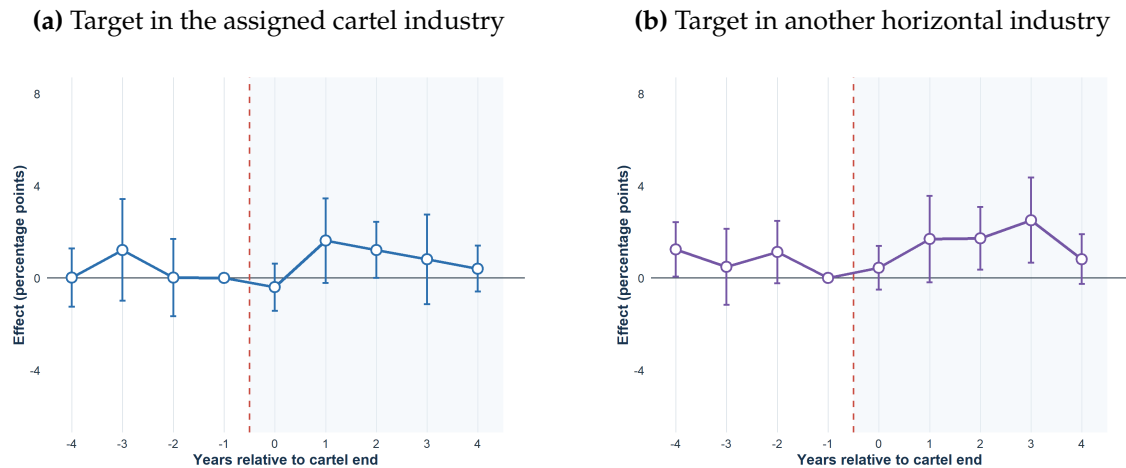


Figure A.3. Where cartel members acquire horizontal minority stakes

Notes. The figure separates acquisitions according to whether the target operates in the four-digit NACE industry assigned to the cartel event. Both panels use outcome-specific pre-event comparisons. Four-digit NACE industries are descriptive proxies and not legal antitrust markets. The reference period is $t = -1$. Bars are 95 percent confidence intervals clustered by firm.

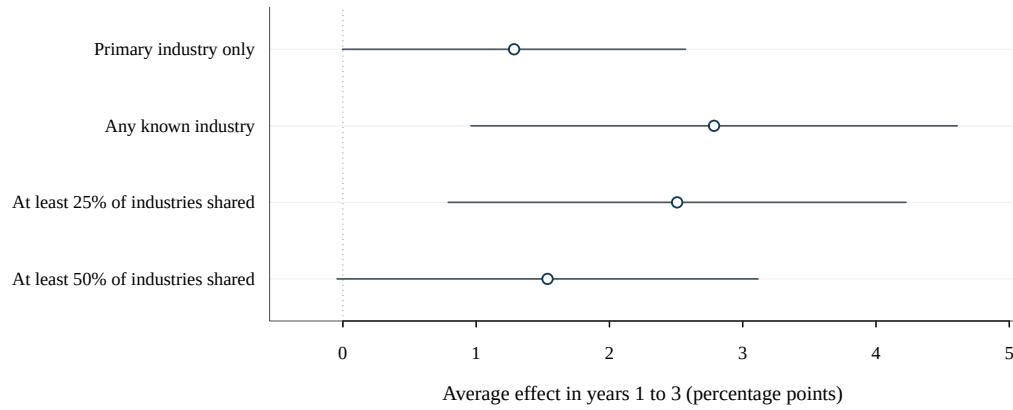


Figure A.4. Acquirer effects under stricter measures of relatedness

Notes. Points are average effects over $t = 1, 2, 3$. The stricter measures divide the number of shared four-digit industries by the total number of distinct industries assigned to the buyer and target. The ratio is calculated separately in SIC, NACE, NAICS, and Bureau van Dijk classifications. Bars are 95 percent confidence intervals clustered by firm.

Table A.5. Outcome and matching robustness

Specification	Effect	95% CI	p -value	Joint pre- p
Primary industries, transaction-wide	1.42	[0.11, 2.73]	0.033	0.408
All known industries, transaction-wide	2.78	[0.96, 4.61]	0.003	0.575
Primary industries, buyer–target pair	1.28	[−0.00, 2.57]	0.050	0.408
All known industries, buyer–target pair	2.78	[0.96, 4.61]	0.003	0.575
At least 25 percent of listed industries shared	2.51	[0.79, 4.23]	0.004	0.540
At least 50 percent of listed industries shared	1.54	[−0.04, 3.11]	0.057	0.346
Broader comparison pool	2.34	[0.16, 4.53]	0.035	0.270
Largest comparison weight limited to 90 percent	2.79	[1.11, 4.46]	0.001	0.458
Largest comparison weight limited to 80 percent	2.84	[1.15, 4.52]	0.001	0.583

Notes. Acquirer mean over $t = 1, 2, 3$, in percentage points, with confidence intervals clustered by firm. The first six rows use the same matched sample. The shared-industry ratio is defined in equation (A.1). The broader comparison draws on firms from progressively wider industries when a close four-digit comparison is unavailable. The final two rows limit the influence of any single comparison firm without using outcomes after the cartel ending.

The random-target benchmark reported in Section 6.4 draws pseudo-targets from the universe of completed minority transactions. For each of the 103 minority deals completed by members during $t = 1, 2, 3$ with an identifiable target country (5 acquirer–target pairs without one are excluded), a pseudo-target is drawn from minority-deal targets in the same completion year and target country and with a similar number of recorded four-digit codes, excluding the actual target and the acquirer itself; 9,999 complete assignments are simulated. Without the code-count stratification, the random benchmarks are 23.4 percent under the main definition and 12.4 percent under primary-only coding, so the conclusion does not depend on that adjustment.

Table A.6 reports the full-acquisition estimate and the destination split; the composition

estimates behind them appear in Table 5 in the main text.

Table A.6. Full acquisitions and the destination of minority investment

Outcome	Effect	95% CI
Any full acquisition	0.12	[−3.47, 3.71]
Stake in cartel-associated industry	1.22	[−0.45, 2.88]
Horizontal stake in other industries	1.97	[0.28, 3.67]
Cartel-associated industry minus other	−0.75	[−3.25, 1.74]

Notes. Acquirer-side effects, averaged over $t = 1, 2, 3$, in percentage points. The full-acquisition estimate uses an outcome-specific comparison and the transaction provider’s historical full-acquisition classification. The remaining rows separate horizontal minority acquisitions by whether the target lies in the four-digit industry attached to the cartel case. Confidence intervals cluster by firm.

A.5 The post-ending transactions

Table A.7 lists the 39 horizontal minority acquisitions completed by members during the first three post-ending years, together with the reported ownership shares and the location of each target relative to the assigned cartel industry. Members are indexed consistently across rows rather than named.

Table A.7. The 39 post-ending horizontal minority acquisitions

Year	Member	Cartel case	Ctry	Stake (%)	Assigned
2003	M1	Thread	GB	16.7	–
2003	M1	Thread	AU	15.8	–
2004	M2	Carglass	RU	44.5	Yes
2004	M1	Thread	AU	7.0	–
2004	M1	Thread	GB	4.4	–
2004	M1	Thread	GB	3.7	–
2005	M3	Professional videotape	JP	10.0 ^a	–
2005	M4	Bathroom fittings & fixtures	KY	4.3 ^a	–
2006	M5	E.ON–GDF	CZ	16.5 ^a	–
2006	M6	Gas insulated switchgear	JP	12.3	–
2006	M5	E.ON–GDF	PL	40.0	Yes
2006	M5	E.ON–GDF	CZ	49.0 ^a	–
2007	M5	E.ON–GDF	RU	47.4	Yes
2008	M7	LCD	TW	39.7 ^a	Yes
2008	M8 ^b	Smart card chips	US	–	Yes
2008	M9	Marine hoses	JP	16.0	–
2009	M10	Envelopes	GB	45.0	–
2009	M7	LCD	JP	–	Yes
2010	M11	Swiss franc interest rate derivatives	JP	5.5	Yes
2010	M11	Swiss franc interest rate derivatives	JP	7.2	Yes
2010	M11	Swiss franc interest rate derivatives	JP	3.8	Yes
2011	M12	Yen interest rate derivatives	US	–	Yes
2011	M13	Power cables	US	–	–
2011	M13 ^b	Power cables	US	–	–
2012	M13 ^b	Power cables	US	–	–
2012	M11	Swiss franc interest rate derivatives	BH	2.7 ^a	Yes

Table A.7 (continued)

Year	Member	Cartel case	Ctry	Stake (%)	Assigned
2012	M14	Power cables	SE	–	–
2012	M12 ^b	Yen interest rate derivatives	GB	–	–
2012	M15	Braking systems; spark plugs	DE	26.2	–
2012	M12	Yen interest rate derivatives	GB	–	–
2012	M12	Yen interest rate derivatives	JP	40.0	Yes
2012	M11	Swiss franc interest rate derivatives	JP	5.2	–
2013	M16	Power exchanges	GB	–	–
2013	M12	Yen interest rate derivatives	US	–	–
2014	M17 ^b	Automotive bearings	SE	–	–
2014	M15 ^b	Braking systems; spark plugs	DE	–	–
2014	M18 ^b	Spark plugs	US	–	–
2014	M19	Car battery recycling	US	–	–
2014	M15	Braking systems; spark plugs	DE	–	–

Notes. Source: Bureau van Dijk's Zephyr database. The table lists the 39 unique horizontal minority acquisitions completed by cartel members during $t = 1, 2, 3$. Members are indexed M1–M19, consistently across rows, rather than named; the 19 indices correspond to the 19 unique acquirers in Table A.4. "Ctry" is the target's headquarters country. "Stake" is the final ownership share where reported; ^a marks deals for which only the acquired share is reported, and "–" indicates that Zephyr records the minority classification without a percentage. ^b marks deals completed by an investor group that includes the member. "Assigned" marks targets operating in the four-digit NACE industry attached to the member's cartel case. Cartel-case names are abbreviated; three transactions by one member are attached to both of its cartel events ending in 2011 (braking systems and spark plugs) and appear once each.

A.6 Design robustness

Figure A.5 varies the event definition and the matching window. The first two panels retain only the earliest or latest eligible cartel event for each firm (average effects of 2.63 and 2.21 percentage points). The third panel retains cartels ending in 2005 or later, selects comparison firms using acquisition activity in $g - 8, \dots, g - 5$ only, and leaves the plotted $g - 4, \dots, g - 2$ coefficients unused in the selection; the post-ending effect in this sample is 4.51 percentage points [0.70, 8.33]. Giving every cartel case the same total treated weight, irrespective of the number of observed member events, yields a mean effect of 5.49 percentage points over $t = 1, 2, 3$ (cartel-clustered 95 percent interval [1.03, 9.95], $p = 0.017$).

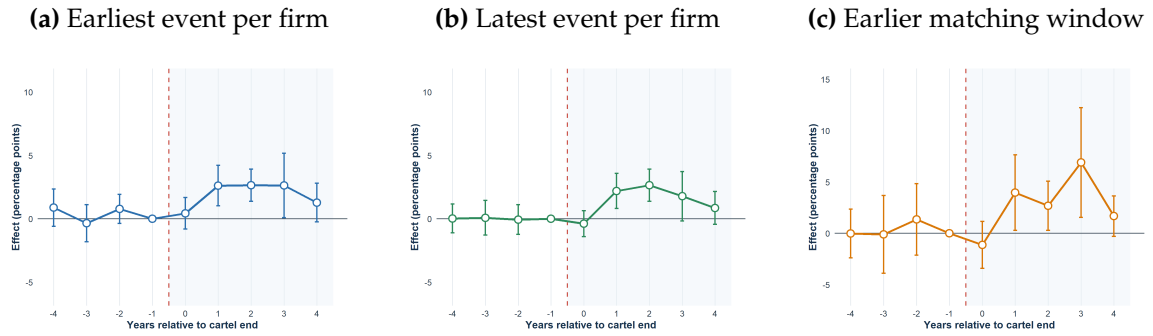


Figure A.5. Alternative event and preperiod constructions

Notes. The first two panels retain only the earliest or latest eligible cartel event for each firm. The third panel retains cartels ending in 2005 or later, matches on acquisition activity in $g - 8, \dots, g - 5$, and leaves the plotted $g - 4, \dots, g - 2$ coefficients unused in control selection. Confidence intervals cluster by firm.

A.7 False early event dates

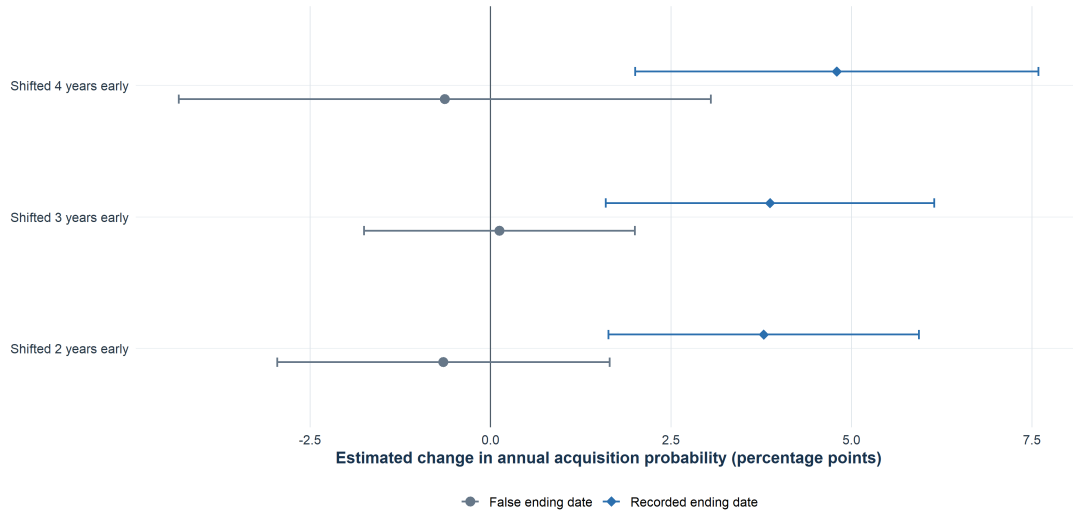


Figure A.6. Timing falsification tests

Notes. Gray circles are effects estimated after shifting cartel endings two, three, or four years earlier. Only pseudo-post years that precede the actual ending enter the falsification window. Blue diamonds are actual-date estimates re-estimated on exactly the same support. Bars are 95 percent confidence intervals clustered by firm.

A.8 Recorded ending and Commission decision timing

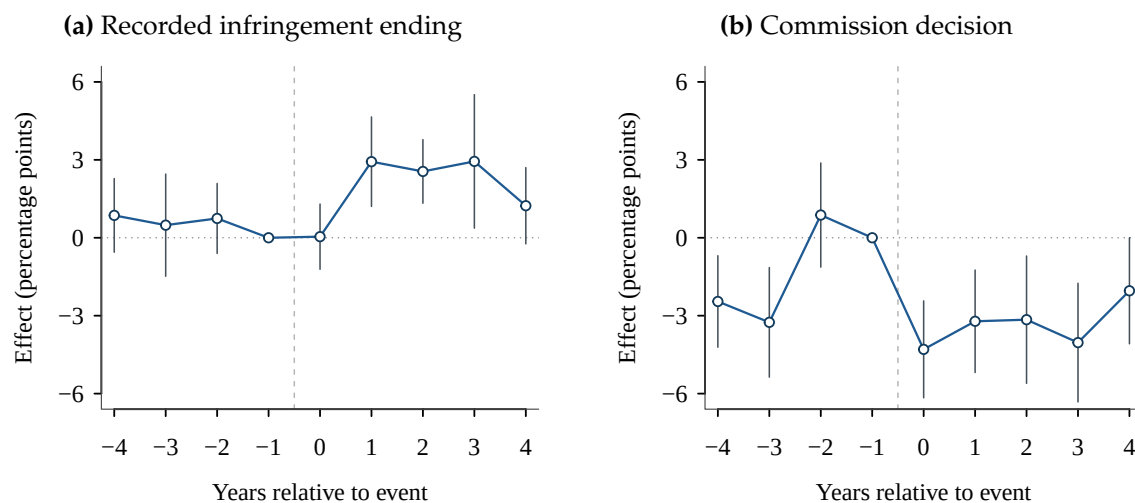


Figure A.7. Acquisition activity around two dates in the enforcement record

Notes. Both panels use the same 244 member events and the same comparison firms. Two events from the Metal Packaging case are excluded because the Commission decision arrived only in 2022, leaving no complete four-year post-decision window in transaction data ending in 2025; the main analysis retains all 246 events. Panel (a) aligns observations on the recorded final infringement year. Panel (b) aligns them on the Commission decision year, which follows the ending by a median of five years. The second panel shows when the response occurs relative to the decision. It should not be interpreted as the causal effect of that decision. Bars are 95 percent confidence intervals clustered by firm.

A.9 Permutation inference

Figure 8 reports two permutation exercises. The first can be run for the 224 matched groups that contain at least two never-named comparison firms. Within each group, one comparison firm is selected with equal probability and assigned the cartel ending, while the actual member is excluded. This produces a benchmark centered on zero and asks whether the member estimate is unusual among otherwise eligible firms. The second exercise uses all 246 member events and randomly reverses the contribution of each cartel. It therefore preserves cartel size, ending year, the matched comparisons, and the common shocks shared by members of the same cartel. Both distributions are based on 999 draws.

A.10 Leave-one-out influence

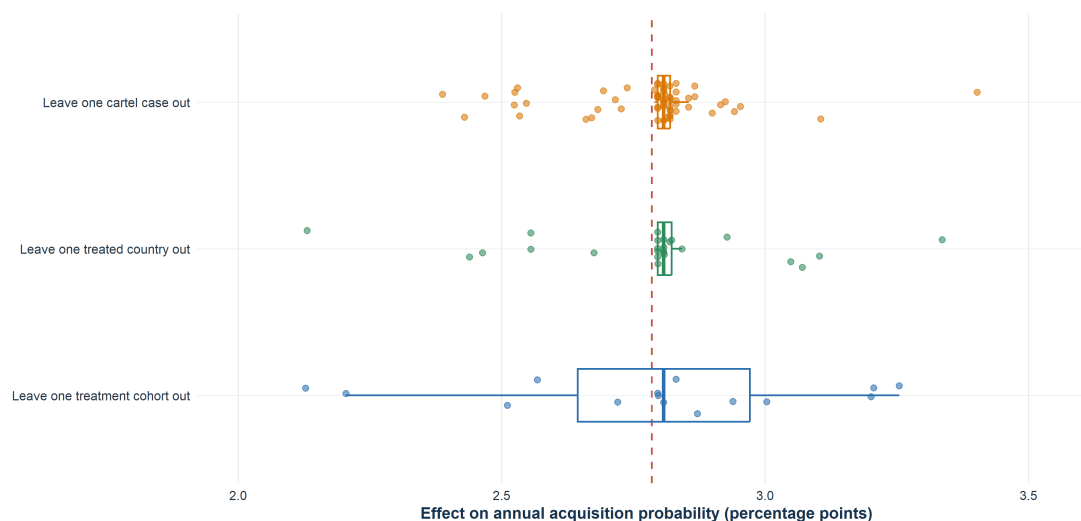


Figure A.8. Leave-one-out estimates

Notes. Each point is the acquirer mean over $t = 1, 2, 3$ from a complete re-estimation that omits one cartel case, one treated country, or one ending cohort together with the attached controls. The dashed line is the full-sample estimate. Boxes summarize each leave-one-out family.

A.11 Cartel-affected industries

The main analysis follows firms named in a Commission decision. A separate design asks whether the same acquisition response appears among all observed firms in an affected NACE4 industry. Overlapping cartels make sector timing ambiguous, so we date each industry once by its first observed cartel ending and once by the final observed exit from cartel activity. The estimates in Table A.8 are small and imprecise under both conventions. Because this design uses a different treatment population and its own matched sample, it is not a direct member-versus-nonmember comparison; the two direct-comparison rows match named members to never-named firms inside the same affected industry.

Table A.8. Named members and cartel-affected industries

Treatment population and event	Role	Firms	Effect	95% CI	Pre- <i>p</i>
Named members, assigned case end	Acquirer	235	2.78	[0.96, 4.61]	0.575
Named members, assigned case end	Target	235	−0.15	[−2.08, 1.78]	0.234
Named members vs. same-sector nonmembers, first ending	Acquirer	43	4.50	[−0.74, 9.73]	–
Named members vs. same-sector nonmembers, final exit	Acquirer	53	2.52	[−2.30, 7.34]	–
All sector firms, first case end	Acquirer	393	0.11	[−0.84, 1.06]	0.660
All sector firms, first case end	Target	393	−0.06	[−1.67, 1.55]	0.937
All sector firms, final cartel exit	Acquirer	495	0.72	[−0.06, 1.51]	0.203
All sector firms, final cartel exit	Target	495	−0.95	[−4.73, 2.83]	0.191

Notes. Effects average $t = 1, 2, 3$ and are percentage-point changes. Each treatment population has its own matched sample. “Firms” counts unique treated firms. The two direct member-versus-nonmember rows match named members to never-named firms inside the same affected NACE4 and use 95 percent intervals clustered by sector case; the first-ending estimate has $p = 0.086$ and the final-exit estimate $p = 0.282$. Other rows use firm-clustered intervals. For the sector-final acquirer estimate, clustering by sector event gives $p = 0.202$.

A.12 Cartel cases represented in the estimation sample

The following catalog links the empirical sample back to the enforcement records. “Reported firms” is the number of companies recorded for the Commission case. “Matched events” counts the member–case observations that survive firm linkage, financial-data requirements, the nine-year event window, and the requirement that a comparable firm can be found.

Table A.9. Cartel cases represented in the estimation sample

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
SAS + Maersk Air	Air transport	European Economic Area	1998	2001	2001	2	1
PO/Copper plumbing tubes	Installation of industrial machinery and equipment	European Economic Area	1988	2001	2004	21	7
Raw Tobacco (ES)	Manufacture of tobacco products	Spain	1996	2001	2004	15	1
Rubber chemicals	Manufacture of other chemical products	European Economic Area	1996	2001	2005	8	3
Thread	Preparation and spinning of textile fibres	European Economic Area	1990	2001	2005	16	1
Industrial bags	Manufacture of plastic packing goods	Germany, Belgium, Luxembourg, Netherlands, France, Spain	1982	2002	2005	25	6

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Raw Tobacco IT	Wholesale of grain, unmanufactured tobacco, seeds and animal feeds	Italy	1995	2002	2005	6	2
Bitumen Nederland	Manufacture of other inorganic basic chemicals	European Economic Area	1994	2002	2006	31	2
Butadiene Rubber / Emulsion Styrene Butadiene Rubber	Manufacture of synthetic rubber in primary forms	European Economic Area	1996	2002	2006	13	1
Methacrylates	Manufacture of plastic plates, sheets, tubes and profiles	European Economic Area	1997	2002	2006	14	3
Bitumen Spain	Manufacture of refined petroleum products	Spain	1991	2002	2007	13	4
Chloroprene Rubber	Manufacture of synthetic rubber in primary forms	European Economic Area	1993	2002	2007	11	1
Professional videotape	Manufacture of consumer electronics	European Economic Area	1999	2002	2007	8	1

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Bananas	Growing of tropical and subtropical fruits	Belgium, Denmark, Germany, France, Greece, Spain, Sweden	2000	2002	2008	8	1
Nitrile Butadiene Rubber	Manufacture of synthetic rubber in primary forms	European Economic Area	2000	2002	2008	4	3
DRAMS	Manufacture of computers and peripheral equipment	European Economic Area	1998	2002	2010	24	2
Pre-stressing steel	Manufacture of basic iron and steel and of ferro-alloys	European Economic Area	1984	2002	2010	36	11
Elevators and escalators	Manufacture of lifting and handling equipment	Netherlands	1995	2003	2007	25	4
Carglass	Manufacture of other parts and accessories for motor vehicles	European Economic Area	1998	2003	2008	18	7

Continued on next page

Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
International Removal Services	Other transportation support activities	Belgium	1984	2003	2008	31	2
Fittings	Installation of industrial machinery and equipment	European Economic Area	1988	2004	2006	33	4
Gas insulated switchgear	Manufacture of electricity distribution and control apparatus	European Economic Area	1988	2004	2007	20	4
Animal Feed Phosphates	Manufacture of chemicals and chemical products	Austria, Belgium, Finland, Denmark, Germany, Hungary, France, Norway, Sweden, UK, Spain, Portugal	1969	2004	2010	13	7

Continued on next page

Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Bathroom fittings & fixtures	Manufacture of ceramic sanitary fixtures	Germany, Austria, Italy, Belgium, France, Netherlands	1992	2004	2010	62	25
CRT glass bulbs	Manufacture of glass and glass products	European Economic Area	1999	2004	2011	4	2
Flat Glass	Manufacture of flat glass	European Economic Area	2004	2005	2007	9	4
Candle waxes	Manufacture of chemicals and chemical products	European Economic Area	1992	2005	2008	28	2
E.On - GdF collusion	Distribution of gaseous fuels through mains	France, Germany	1980	2005	2009	3	1
Consumer detergents	Manufacture of soap and detergents, cleaning and polishing preparations	Belgium, France, Germany, Greece, Italy, Portugal, Spain, Netherlands	2002	2005	2011	5	1
Exotic fruit	Growing of perennial crops	Greece, Italy, Portugal	2004	2005	2011	6	3

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Smart card chips	Manufacture of electronic components	European Economic Area	2003	2005	2014	9	2
LCD	Manufacture of computers and peripheral equipment	European Economic Area	2001	2006	2010	8	2
TV and computer monitor tubes (Colour Picture Tubes)	Manufacture of computer, electronic and optical products	Worldwide	1996	2006	2012	12	1
Automotive Wire Harnesses (Renault W52/95-Platform)	Manufacture of electrical and electronic equipment for motor vehicles	European Economic Area	2000	2006	2013	10	2
Calcium carbide and magnesium based reagents	Manufacture of other inorganic basic chemicals	European Economic Area	2004	2007	2009	15	3
Marine hoses	Manufacture of rubber and plastic products	European Economic Area	1986	2007	2009	11	6

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Refrigeration compressors	Manufacture of other pumps and compressors	European Economic Area	2004	2007	2011	10	3
Freight Forwarding (PSS)	Freight air transport	Worldwide	2002	2007	2012	44	2
Mountings for windows and window-doors	Manufacture of locks and hinges	European Economic Area	1999	2007	2012	11	2
Lighting systems	Manufacture of electric lighting equipment	European Economic Area	2004	2007	2017	6	2
Water management products	Manufacture of tanks, reservoirs and containers of metal	European Economic Area	2006	2008	2012	7	2
Euro Interest Rate Derivatives (EIRD)	Trusts, funds and similar financial entities	European Economic Area	2005	2008	2013	19	2
Envelopes	Manufacture of paper and paper products	European Economic Area	2003	2008	2014	16	8
Optical Disc Drives	Manufacture of magnetic and optical media	Worldwide	2004	2008	2015	15	1

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Retail Food Packaging - France	Manufacture of plastic packing goods	European Economic Area	2000	2008	2015	40	6
Shrimps	Wholesale of other food, including fish, crustaceans and molluscs	European Economic Area	2000	2009	2013	10	1
Power Cables	Manufacture of electricity distribution and control apparatus	European Economic Area	1999	2009	2014	26	9
Swiss Franc Interest Rate Derivatives (Bid Ask Spread Infringement -CHF Libor)	Other activities auxiliary to financial services, except insurance and pension funding	European Economic Area	2007	2009	2014	8	1
Thermal Systems	Manufacture of electrical and electronic equipment for motor vehicles	European Economic Area	2004	2009	2017	14	7
Yen Interest Rate Derivatives (YIRD)	Trusts, funds and similar financial entities	European Economic Area	2007	2010	2013	15	3

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Polyurethane Foam	Manufacture of furniture	European Economic Area	2005	2010	2014	30	14
Occupant Safety Systems supplied to Japanese Car Manufacturers	Manufacture of other parts and accessories for motor vehicles	European Economic Area	2004	2010	2017	6	3
Automotive Bearings	Manufacture of bearings, gears, gearing and driving elements	European Economic Area	2004	2011	2014	19	12
Trucks	Manufacture of motor vehicles	European Economic Area	1997	2011	2016	18	3
Braking Systems	Manufacture of other parts and accessories for motor vehicles	European Economic Area	2007	2011	2018	7	3
Spark Plugs	Manufacture of electrical and electronic equipment for motor vehicles	European Economic Area	2000	2011	2018	4	5

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Occupant Safety Systems	Manufacture of other parts and accessories for motor vehicles	European Economic Area	2007	2011	2019	7	2
Mushrooms	Growing of vegetables and melons, roots and tubers	European Economic Area	2010	2012	2014	10	2
Power exchanges	Trade of electricity	Worldwide	2011	2012	2014	2	2
Block trains	Freight rail transport	European Economic Area	2004	2012	2015	10	3
Car battery recycling	Recovery of sorted materials	Germany, Belgium, Netherlands, France	2009	2012	2017	11	3
Capacitors	Manufacture of electronic components	European Economic Area	1998	2012	2018	15	4
Closure systems	Manufacture of other parts and accessories for motor vehicles	European Economic Area	2009	2012	2020	10	2

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Table A.9 continued

Case	Affected sector	Affected geography	Start	End	Decision	Reported firms	Matched events
Forex (Three Way Banana Split - Essex Express - Sterling Lads)	Other financial service activities, except insurance and pension funding n.e.c.	European Economic Area	2007	2013	2019	20	3
Canned vegetables	Other processing and preserving of fruit and vegetables	European Economic Area	2000	2013	2021	11	1
Metal packaging	Manufacture of light metal packaging	Germany	2011	2014	2022	7	2
SSA Bonds	Activities auxiliary to financial services, except insurance and pension funding	European Economic Area	2010	2015	2021	9	1

Notes. Start and end are the first and last recorded infringement years in the Commission case database. Decision is the European Commission decision year. Reported firms is the database count of companies named for the case. Matched events is the number of member-case observations that enter the estimation sample. A firm can contribute to more than one case.